

SatFedGuard: Semi-Supervised Federated Contrastive Learning With RL-Assisted Bidirectional Distillation for Anomaly Traffic Detection in Satellite Networks

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Abstract—Federated learning-based intrusion detection methods for satellite networks enable model training without sharing local data, thereby ensuring network security while significantly reducing communication overhead. However, due to the difficulty of obtaining large-scale high-quality labeled data in satellite environments, a key challenge lies in how to train intrusion detection models using abundant unlabeled traffic data. We propose SatFedGuard, a semi-supervised federated contrastive learning approach for anomaly traffic detection in satellite networks. SatFedGuard effectively integrates unlabeled in-orbit data with labeled data from ground stations for model training. First, it models the unlabeled satellite traffic data using a contrastive learning framework. To address the challenge of non-IID data distribution, an attention-based dual-path aggregation strategy is designed to generate personalized models for each satellite by leveraging model similarities. Then, a bidirectional multi-granularity distillation method between larger and smaller models is implemented, where reinforcement learning is employed to optimize the weights of different loss terms dynamically. Experiments on two satellite network traffic datasets under non-IID settings demonstrate that the proposed method significantly improves anomaly detection performance while reducing dependence on in-orbit labeled data, achieving F1-Scores of 93.38% ($\uparrow 11.63\%$) and 99.80% ($\uparrow 8.72\%$), respectively.

Index Terms—Federated learning, satellite network, intrusion detection, semi-supervised learning, edge-cloud collaboration.

I. INTRODUCTION

SATELLITE network refers to a global space-based communication infrastructure system constructed using

communication satellites distributed in Earth's orbit [1], [2], [3]. With the continuous advancement of sixth-generation mobile communication technology (6G), the future of global communications is gradually evolving toward an integrated space-air-ground network architecture [4], [5], [6], [7]. In this context, communication networks centered around large-scale Low Earth Orbit (LEO) satellites are widely regarded as the key support for achieving seamless global connectivity. Due to their lower orbital altitude, LEO satellites have significant advantages in terms of communication latency, cost control, and spectrum utilization, making them well-suited to collaboratively build high-capacity, low-latency, and highly reliable communication systems with terrestrial networks [8]. At present, satellite networks not only play a crucial role in promoting global Internet access but also demonstrate important strategic value in scientific research [9], national defense [10] and security [11], disaster response [12], intelligent transportation [13], and other fields. Governments and technology enterprises around the world are investing heavily and launching large-scale LEO satellite constellation construction plans, striving to seize the initiative in future global communication competition [14]. Satellite networks have gradually become one of the indispensable infrastructures of modern information society [15], [16].

Unlike traditional terrestrial networks, satellite networks are characterized by high openness [17] and dynamical topologies [18]. Due to these features, satellite networks face higher security risks, and network attacks launched by malicious users are more likely to affect the normal operation of the network on a large scale [19]. Therefore, deploying effective malicious traffic detection systems onboard satellites is of great significance [20], [21]. However, the vast volume of data generated by mega-constellations presents significant challenges to conventional centralized processing paradigms. These challenges include high latency in satellite-to-ground communications, constrained bandwidth resources, and critical concerns regarding data ownership and privacy preservation [22]. Furthermore, the unique environmental constraints of satellite networks, such as intermittent connectivity caused by high-speed mobility, high propagation delays in space-ground links, and bursty link conditions, further hinder the real-time backhaul of raw traffic data for centralized analysis.

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With the steady improvement of onboard computing capabilities and the strong demand for space-native intelligence, satellite-based federated learning has become an effective solution to this problem [23]. Federated learning is particularly necessary for satellite anomaly detection because it enables satellites to collaboratively build a globally optimal model without exchanging local data, thereby inherently addressing bandwidth bottlenecks and privacy concerns. On this basis, each satellite node only needs to periodically upload model parameters or gradient information, thus avoiding the high communication overhead of raw data transmission over downlink channels and alleviating system bottlenecks caused by limited bandwidth and high latency [24]. Moreover, its decentralized nature provides essential fault tolerance; even if certain nodes experience temporary dropouts due to intermittent link interruptions or bursty conditions, the global training process can remain stable [25], which is a decisive advantage over centralized paradigms in volatile space environments.

However, existing satellite-based federated learning approaches remain limited in effectively addressing two major challenges: insufficient data annotation and data heterogeneity. First, due to the difficulty of obtaining large-scale labeled data in orbit and the limited per-client data availability on individual LEO satellites [26], traditional supervised learning is difficult to implement effectively. While unsupervised learning relies solely on intrinsic features, it lacks clear optimization direction and performs unstably in complex environments [27], [28], [29]. Second, satellite networks are characterized by Non-IID data distribution. Due to varying orbital positions and geographic coverage, traffic patterns across different satellites exhibit significant diversity. A single global model aggregated via traditional Federated Averaging (FedAvg) often fails to adapt to these diverse local distributions, leading to poor robustness. Therefore, how to fully leverage the federated learning paradigm in space environments where labeled data is extremely scarce and data distribution is heterogeneous remains a key challenge.

To address the above problems, this paper proposes SatFedGuard, an anomaly traffic detection method for satellite network scenarios. It is based on semi-supervised federated contrastive learning, attention-based personalized aggregation, and reinforcement learning-assisted transfer learning mechanisms. It is capable of fully exploiting the rich information of unlabeled data collected in orbit, while also using a small amount of labeled samples gathered and annotated by ground stations to guide and optimize the model. Specifically, to tackle the Non-IID challenge, we design a dual-path aggregation strategy to generate personalized models for each satellite. Furthermore, to enhance knowledge transfer efficiency, we employ a reinforcement learning-assisted bidirectional distillation framework. The proposed method is suitable for model training in onboard anomaly traffic detection scenarios, where satellites lack local labeled data, while ground stations are able to obtain and annotate a small amount of historical data. The goal of this method is to make efficient use of limited ground-labeled resources to achieve efficient and robust detection of anomalous traffic on satellites. The main contributions of this paper are as follows:

- A satellite-ground collaborative learning architecture with models of different scales is proposed for anomaly traffic detection in satellite networks. This architecture first migrates semi-supervised federated contrastive learning to the satellite network, constructs a multi-to-one star topology between satellites and ground stations, utilizes unlabeled satellite data and labeled ground station data for model training, and synchronizes model parameters via satellite-ground links, thereby avoiding direct transmission of traffic data. This collaborative design effectively alleviates the resource constraints of satellites while leveraging the powerful computing capabilities of ground stations.
- A satellite-based semi-supervised federated contrastive learning method is proposed. By executing the Barlow Twins algorithm on satellites for self-supervised learning, the method fully leverages in-orbit data. Meanwhile, ground stations perform fine-tuning with a small amount of labeled data and conduct bidirectional distillation with a larger model, achieving better performance than traditional algorithms and effectively addressing the difficulty of traffic labeling in satellite networks.
- An attention-based dual-path aggregation strategy and a reinforcement learning-assisted transfer learning mechanism are proposed. To address the data heterogeneity in satellite networks, we design an attention-based aggregation method that concurrently generates a global model and personalized models, significantly enhancing the model's adaptability and robustness. Furthermore, we implement a multi-granularity bidirectional distillation framework assisted by the PPO (Proximal Policy Optimization) algorithm. This acts as an efficient transfer learning mechanism, dynamically optimizing the knowledge flow between the server's large model and the satellites' personalized models.

The paper is organized as follows. Section II reviews the related work. Section III describes the system model. Section IV presents the proposed architecture and methodology in detail. Section V discusses the experimental results. Finally, Section VI concludes the paper and outlines future work.

II. RELATED WORKS

A. Intrusion Detection in Satellite Networks

Intrusion detection in satellite networks has increasingly adopted intelligent methods such as machine learning and deep learning in recent years, in order to address the network's complex characteristics, including high dynamism, high latency, and limited resources. Pacheco et al. [30] proposed a hierarchical classification framework that applies machine learning and deep learning techniques to classify heterogeneous Internet traffic—such as encrypted, unencrypted, and tunneled data—in satellite communications, thereby improving QoS. Azar et al. [31] introduced SAT-IDSs, which leverage a random forest-based Sequential Forward Selection (RF-SFS) method to identify key features, and combine them with models such as Random Forest, Artificial Neural Networks (ANN), Long

Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) to enhance the accuracy and efficiency of malicious traffic detection in both satellite and ground networks. Driouch et al. [32] proposed CANSat-IDS, an adaptive distributed intrusion detection system that integrates ANN and K-SVM. Onboard, it performs real-time detection of DoS and fuzzing attacks based on timestamps and CAN-IDs; on the ground, it analyzes data fields to detect replay attacks. The distributed architecture of “real-time detection–offline training–firmware update” balances detection performance with the resource limitations of satellites. Xu et al. [33] addressed the challenge of acquiring labeled traffic data in SAGIN by enhancing the HAST-NAD network model and introducing a skip connection mechanism to transition from supervised to semi-supervised learning. They also built a horizontal federated learning architecture centered on ground gateways with high-orbit satellites as participants, ensuring data security while conserving bandwidth. He et al. [34] utilized federated learning to collaboratively train a differentially private conditional diffusion model, which generates data representing the global traffic distribution. Each node then combines this synthetic data with local data to train intrusion detection models, enabling online detection of network intrusions.

Although the above approaches have made notable progress in improving the accuracy of intrusion detection in satellite networks, they still face two major limitations. First, centralized learning methods [30], [31], [32] overlook the substantial communication overhead caused by the downlink transmission of raw data—an issue particularly significant in large-scale LEO satellite environments. Second, although some studies [33], [34] have adopted distributed learning architectures and advanced related technologies, the high cost of in-orbit annotation for satellite data poses challenges to their widespread application.

B. Intrusion Detection Based on Semi-Supervised Learning

In the field of intrusion detection, semi-supervised learning is gradually becoming a mainstream solution to address the scarcity of labeled data. Such methods aim to efficiently utilize massive amounts of unlabeled data by leveraging semi-supervised learning techniques. Jiang et al. [35] proposed an incremental dynamic weighted semi-supervised learning approach that integrates a block-based strategy with fuzzy rule-based classification, enabling effective model updates using limited labeled samples and abundant unlabeled data. Dang et al. [36] reformulated unsupervised anomaly detection as a proxy supervised task by projecting normal samples into a Gaussian latent space and generating pseudo-anomalies, achieving efficient detection without prior anomaly knowledge. Zhao et al. [37] further enhanced semi-supervised detection by combining collaborative interpolation with pseudo-labeling, where specialized traffic data augmentation and interpolation mechanisms were employed to refine decision boundaries under extremely scarce labeling conditions. However, these methods are generally designed for centralized environments, while federated semi-supervised learning is better suited to distributed settings where data are

decentralized. Zhao et al. [38] proposed SSFL, an intrusion detection method based on semi-supervised federated learning and knowledge distillation. In this approach, clients first train classifiers using local labeled data, then train discriminators to filter prediction results. Hard labels are uploaded after screening, and the server determines global hard labels through a voting mechanism. Clients perform distillation training using global hard labels, thereby reducing communication overhead while improving detection performance. Wang et al. [39] proposed a federated semi-supervised learning-based network traffic classification method. This approach labels traffic on edge devices using DPI and DNS, and utilizes a convolutional autoencoder-based semi-supervised classification model composed of an encoder, decoder, and classifier. Through alternating training, the model extracts high-dimensional features from unlabeled samples and performs classification on labeled samples, reducing reliance on labeled data. The global model is updated via federated learning to enhance traffic classification performance while preserving data privacy. Aouedi et al. [40] proposed a federated semi-supervised learning approach for attack detection in industrial IoT. An autoencoder is used on local devices to learn low-dimensional features from unlabeled data, and after aggregating the global model through federated learning, a fully connected layer is added in the cloud to perform supervised learning using public labeled data. Li and Li et al. [41] introduced a security threat detection framework named FedUPS based on semi-supervised federated learning. It uses CNNs for feature extraction and applies uncertainty-aware pseudo-label selection to generate reliable pseudo-labels for unlabeled data. A dynamic weighted aggregation method is designed to optimize model updates, addressing the issues of data privacy and unlabeled data utilization in distributed power control systems.

The above methods demonstrate the great potential of federated semi-supervised learning in the field of intrusion detection, but they still face unique challenges in the context of satellite networks. Although these methods reduce dependence on labeled data, some [35], [36], [37] rely on centralized environments and are difficult to adapt to satellite scenarios, while others [38], [39], [41] still require a certain amount of in-orbit labeled data, which is difficult to obtain in satellite scenarios. Moreover, current solutions [38], [39], [40] generally adopt a homogeneous model structure between satellites and ground stations—i.e., both sides use lightweight models to match the limited computational capacity of satellites. This restricts the full utilization of ground-labeled data, thereby affecting detection accuracy. To provide a clearer comparison, we summarize the specific methodologies and limitations of the aforementioned works in Table I.

In this paper, we propose a contrastive learning-based semi-supervised federated learning method to address the lack of labeled onboard data in satellite network intrusion detection. Furthermore, to enhance the performance of edge-side models, we introduce a reinforcement learning-assisted multi-granularity bidirectional distillation strategy between larger and smaller models, which promotes knowledge transfer between models.

TABLE I
COMPARISON OF RELATED WORKS: SPECIFIC METHODOLOGIES AND LIMITATIONS

Reference	Specific Methodology	Limitation
Pacheco et al. [31]	Proposed a hierarchical framework using ML/DL to classify heterogeneous traffic for QoS improvement.	Relies on centralized learning which requires transmitting raw traffic data to the ground, consuming limited satellite bandwidth.
Azar et al. [32]	Introduced SAT-IDSs using RF-SFS for feature selection and hybrid models for malicious traffic detection.	The centralized processing approach necessitates raw data down-link, leading to substantial communication costs in LEO networks.
Driouch et al. [33]	Proposed CANSat-IDS, a distributed system. Uses onboard real-time detection for DoS/fuzzing and ground offline training.	Despite the distributed split, the offline training and firmware update loop still rely on significant data transmission between space and ground.
Xu et al. [34]	Enhanced the HAST-NAD model with skip connections and established a horizontal federated learning architecture with ground gateways.	The training process relies on acquiring labeled traffic data in the satellite environment, which is difficult and expensive to obtain.
He et al. [35]	Utilized federated learning with a differentially private conditional diffusion model to generate synthetic global traffic data for training.	Although synthetic data is used, the model still implicitly relies on the quality of in-orbit data distributions and labels, which are challenging to acquire.
Jiang et al. [36]	Proposed incremental dynamic weighted SSL combining block-based strategies with fuzzy rule-based classification for model updates.	The method is designed for centralized environments and lacks the adaptation required for the distributed, decentralized nature of satellite networks.
Dang et al. [37]	Reformulated unsupervised anomaly detection as a proxy supervised task by projecting samples into a Gaussian latent space.	The approach assumes a centralized data availability and is not tailored for the distributed constraints of satellite constellations.
Zhao et al. [38]	Combined collaborative interpolation with pseudo-labeling, using traffic augmentation to refine decision boundaries.	Focuses on centralized semi-supervised learning and does not address the data isolation and privacy issues in distributed satellite scenarios.
Zhao et al. [39]	Proposed SSFL using knowledge distillation where clients train discriminators and the server determines global hard labels via voting.	Uses homogeneous lightweight models that fail to utilize ground computing power; still requires a portion of in-orbit labeled data.
Wang et al. [40]	Proposed Fed-SSL using a convolutional autoencoder; edge devices label traffic using DPI and DNS for semi-supervised training.	Relies on homogeneous model structures between space and ground; dependent on DPI/DNS for labeling which may not always be available or accurate.
Aouedi et al. [41]	Used autoencoders for local feature learning and a cloud-based fully connected layer for supervised learning with public data.	Adopts a model structure that matches limited satellite capacity, restricting the potential to leverage stronger ground-based computational resources.
Li et al. [42]	Introduced FedUPS using CNN and Uncertainty-aware Pseudo-label Selection with a dynamic weighted aggregation method.	Still requires a certain amount of in-orbit labeled data for effective pseudo-labeling; limited by homogeneous model constraints.

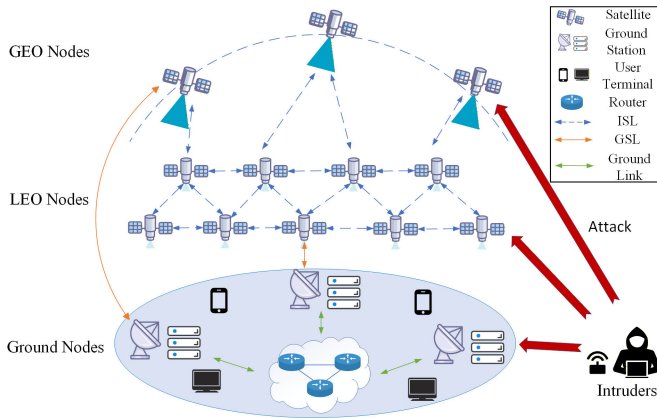


Fig. 1. Space-ground integrated network architecture diagram.

III. OVERVIEW OF THE SYSTEM MODEL

The satellite network illustrated in Figure 1 considers a generic Space-Air-Ground Integrated Network (SAGIN) architecture. This hierarchical design integrates LEO satellites, geostationary Earth orbit (GEO) satellites, and ground nodes to leverage their complementary strengths—combining GEO’s

stable coverage with LEO’s high-speed access. Satellites operate periodically along predetermined orbits, providing dynamic and seamless coverage of ground nodes [42]. Satellites connect to ground nodes via satellite-to-ground links, while inter-satellite communication is enabled through inter-satellite links [43]. Due to their unique orbital characteristics, GEO satellites have ample energy supply and longer design lifespans, allowing them to integrate more advanced and complex computing and communication capabilities [44]. In this work, GEO satellites are responsible for data aggregation, in-orbit model training, and communication with ground stations for uploading and downloading models. LEO satellites, characterized by low latency and high bandwidth, have limited onboard resources. In this framework, they are mainly responsible for collecting and preprocessing traffic data, which is then transmitted to the GEO satellites. The ground nodes, consisting of ground gateway stations, routers, and user terminals, are equipped with ample computing resources and storage capabilities. They transmit data via terrestrial networks, receive models uploaded by GEO satellites, and send the updated global model back to GEO. Ground nodes are also responsible for collecting and labeling a small portion of satellite data and deploying high-performance larger models.

The threat model in this paper considers a range of adversarial attacks targeting the space segment. Satellite systems are capable of communicating not only with ground stations but also with various other devices, such as user terminals and routers, enabling direct data exchange. We assume that the adversary possesses advanced penetration capabilities, allowing them to control and exploit a large number of ground-based network routers and user terminal devices. Leveraging these compromised devices, adversaries can directly target the space segment—such as in-orbit satellites—through various attack vectors, including malicious command injection, signal spoofing, and denial-of-service (DoS) attacks. Additionally, by compromising critical ground infrastructure (e.g., ground control stations), they can perform more covert and sophisticated indirect attacks. Such attacks can result in precise disruption of satellite nodes, theft of sensitive data, or even functional failure and permanent paralysis, posing a severe threat to the stability and security of the entire space system. Therefore, to effectively counter the growing cyber threats in satellite networks, there is an urgent need to develop a robust intrusion detection method specifically designed for satellite environments. By deploying this method across various nodes—especially those in the space segment—it becomes possible to achieve accurate detection of anomalous traffic in satellite networks.

IV. METHODS

Traditional anomaly traffic detection methods in satellite networks typically rely on collecting onboard traffic data and transmitting it via the downlink to ground gateway stations. After model training based on historical data on the ground, the trained model is then deployed back to the satellites. However, with the development of large-scale low Earth orbit (LEO) satellite networks and the surge in network traffic, the reliance on downlink resources in traditional schemes has increasingly intensified. To reduce the consumption of precious downlink bandwidth, the concept of space-native intelligence has gradually emerged, promoting the direct use of local data onboard satellites for training and updating AI models.

To address challenges such as limited computing resources and difficulties in data labeling during in-orbit training, this paper proposes a satellite-ground co-evolutionary approach for anomaly traffic detection. This approach integrates federated semi-supervised contrastive learning with reinforcement learning-assisted bidirectional knowledge distillation strategy, aiming to build an efficient, intelligent anomaly detection model with in-orbit adaptive capabilities, thereby enhancing in-orbit detection performance of satellite networks.

A. System Architecture Design

To enable efficient in-orbit model training under label-scarce conditions, while minimizing reliance on bottleneck-constrained downlink resources, this paper proposes SatFedGuard, a collaborative satellite-ground federated learning architecture designed to facilitate a closed-loop evolution of knowledge. By decoupling the processes of feature extraction, model aggregation, and knowledge transfer,

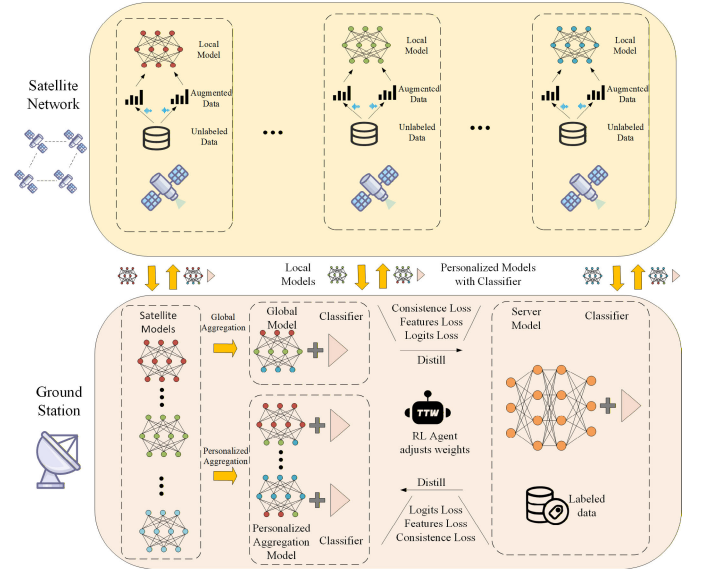


Fig. 2. Overall system architecture diagram.

the framework systematically addresses the unique constraints of satellite environments—such as label scarcity and Non-IID traffic patterns—while ensuring high-fidelity representation learning. As illustrated in Fig. 2, the architecture coordinates distributed satellite nodes and a central ground server to achieve a synergistic balance between onboard computational efficiency and global model generalization.

The operational logic of SatFedGuard is realized through four synergistic functional modules. At the satellite tier, the system employs a self-supervised feature extraction module based on the Barlow Twins algorithm, enabling satellites to learn robust latent representations from local unlabeled traffic without manual annotations. Upon transmission to the ground, an attention-based dual-path aggregation strategy is utilized to concurrently generate a global model and dedicated personalized models by calculating inter-model similarities. Subsequently, a multi-granularity bidirectional knowledge distillation framework is established to facilitate efficient knowledge flow. By aligning knowledge at distinct feature levels, forward distillation integrates general knowledge into the server model, while backward distillation transfers refined domain knowledge back to the personalized models. Finally, to enhance the knowledge transfer efficiency during the distillation process, a reinforcement learning agent based on the PPO algorithm is introduced. This agent autonomously perceives the training state and dynamically adjusts the importance weights of the aforementioned multi-granularity distillation losses, thereby ensuring the stability and efficiency of the system evolution.

B. Contrastive Learning-Based Semi-Supervised Federated Learning Method

To cope with the scarcity of labeled onboard data, we adopt a semi-supervised federated learning framework. On satellites, a contrastive learning algorithm is used for self-supervised learning based on unlabeled data, while supervised learning is performed at the ground station using labeled data.

To cope with the scarcity of labeled onboard data, we adopt a semi-supervised federated learning framework. On satellites, a contrastive learning algorithm is used for self-supervised learning based on unlabeled data to update the local model, while supervised learning is performed at the ground station using labeled data to update the server model.

Barlow Twins is a self-supervised learning algorithm that does not require negative samples. Its goal is to learn robust data representations without relying on manually labeled data. It generates two randomly augmented views from the same input sample, feeds them into twin networks with shared weights to obtain two sets of feature vectors, and optimizes their cross-correlation matrix. The optimization forces the diagonal elements to approach 1 to maintain feature consistency of the same sample under different distortions, while pushing the off-diagonal elements toward 0 to reduce feature redundancy. This avoids the dependency on negative samples typical in traditional contrastive learning. Its optimization objective is as follows:

$$\mathcal{L}_{BT} \triangleq \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{j \neq i} C_{ij}^2 \quad (1)$$

where λ is a scalar constant used to balance the importance between the first and second terms in the loss function, and C is the cross-correlation matrix of the outputs from two identical networks over the batch dimension, computed by the following formula:

$$C_{ij} \triangleq \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}} \quad (2)$$

where z^A and z^B represent the two augmented embedding batches, respectively. The index b denotes the sample in the batch, while i and j indicate the dimensions of the network output vectors. C is a square matrix whose size equals the output dimension of the network, and its values range between -1 and 1 , representing complete negative to complete positive correlation.

In the proposed method, local training on satellites adopts the Barlow Twins algorithm. First, local data are augmented using techniques such as local window jittering, adaptive Gaussian noise addition, block-wise masking, and channel shuffling to obtain two enhanced embeddings. Then, the cross-correlation matrix between them is computed using Eq. (2), and the Barlow Twins loss defined in Eq. (1) is minimized. After multiple rounds of local training, each satellite obtains a local model ϕ_r^k , which is then sent to the ground station for aggregation. Aggregation employs an attention-based dual-path algorithm to simultaneously derive a global model and a personalized model. The global model functions to transfer general knowledge from individual satellites to the server model, whereas the personalized model acquires deep feature knowledge transmitted by the server model.

The ground station employs a DenseNet network [45] as the server model, which serves as a larger-scale encoder, and uses a neural network composed of a fully connected layer as the classifier. It performs supervised learning using the labeled data available at the server, optimizing both the encoder and

classifier. The global model distills the knowledge learned by the satellites into server model, enabling it to learn better feature representations and mitigate the challenge of insufficient labeled data. In supervised learning, the model is typically trained by minimizing the cross-entropy loss function, which is defined as follows:

$$\mathcal{L}_s = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (3)$$

where C denotes the number of classes, y_i is the one-hot encoding of the true label, and $\hat{y}_i$ is the predicted probability of class i produced by the model. The final loss function is defined as follows:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_s + (1 - \alpha) \mathcal{L}_d \quad (4)$$

where $\mathcal{L}_s$ denotes the cross-entropy loss calculated from supervised learning, and α represents the weight of the supervised learning loss.

Due to the limited computing resources onboard satellites, directly deploying the larger model from the ground station onto satellites is impractical, as it would hinder the normal operation and real-time detection of the edge-side models. Therefore, this paper once again leverages knowledge distillation to transfer the knowledge learned by the larger ground model back to the global model, ensuring that the detection model remains lightweight while maintaining high performance.

To achieve this goal, a small classifier is appended to the global model and personalized models on the server. Supervised learning and multi-granularity knowledge distillation are then performed on the combined model using labeled data, with the corresponding computation given in Eq. (4).

After training, the global model is updated using exponential moving average (EMA), where a weighted average of the current global model and the EMA model from the previous round is computed [46]. This approach ensures that model weights decay exponentially over time, giving more importance to recent models while still preserving historical knowledge to some extent. This helps to prevent drastic oscillations during training and improves the stability of model convergence. The model update is defined by the following formula:

$$\phi_{EMA}^r = \begin{cases} \phi^r, & t = 0 \\ \beta \phi^r + (1 - \beta) \phi_{EMA}^r, & t > 0 \end{cases} \quad (5)$$

The updated model will be transmitted to the satellites via the uplink for the next round of training, continuing until convergence is achieved. Ultimately, this process results in a lightweight semi-supervised anomaly traffic detection model based on satellite-ground collaborative evolution. The complete process is shown in Algorithm 1.

C. Attention-Based Dual-Path Aggregation Method

In satellite networks, satellites operate in distinct orbital positions and cover diverse geographic regions, leading to significant heterogeneity in the collected traffic characteristics. Consequently, a global model generated via the standard FedAvg algorithm is often ill-suited to simultaneously adapt

Algorithm 1 Semi-Supervised Federated Learning Method Based on Contrastive Learning

Input : Communication round T ;
 Satellite local epoch E_s ;
 Server local epoch E_{srv} ;
 Number of satellites N ;
 Satellite Set S ;

Output: Trained personalized models $\{\phi_{pers}^i\}_{i=1}^N$ and server model ϕ_{srv}^0

- 1 Initialize server model ϕ_{srv}^0 , server classifier ψ_{srv}^0 , and local models $\{\phi_k^0\}_{k=1}^N$;
- 2 **for** each round $r = 1, 2, \dots, T$ **do**
- 3 Randomly select n satellites $S_r \subseteq S$;
- 4 **for** each satellite $k \in S_r$ **in parallel do**
- 5 $B_k^{unlabel} \leftarrow$ split $D_k^{unlabel}$ into batches of size B ;
- 6 **for** each local epoch $e = 1, 2, \dots, E_s$ **do**
- 7 **for** batch $b \in B_k^{unlabel}$ **do**
- 8 $z_a, z_b \leftarrow \text{Augment}(b)$;
- 9 $C_{ij} \triangleq \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}}$;
- 10 $L_{BT} \triangleq \sum_i (1 - C_{ii})^2 + \lambda \sum_i \sum_{j \neq i} C_{ij}^2$;
- 11 $\phi_k^r = \phi_k^r - \eta \nabla L_{BT}$
- 12 Transmit local model ϕ_k^r to server;
- 13 $\phi_{global}^r \leftarrow \sum_{k \in S_r} \frac{n_k}{m_r} \phi_k^r$
- 14 $Sim_{i,j} \leftarrow \text{Cosine}(\phi_i^r, \phi_j^r)$;
- 15 $\alpha_{i,j} \leftarrow \text{Softmax}(Sim_{i,j})$;
- 16 $\phi_{pers}^{r,i} \leftarrow \sum_{j \in S_r} \alpha_{i,j} \cdot \phi_j^r$
- 17 $(\phi_{srv}^{r+1}, \psi_{srv}^{r+1}), \{(\phi_{pers}^{r+1,i}, \psi_{pers}^{r+1,i})\} = \text{Algorithm2}()$;
- 18 Distribute $(\phi_{pers}^{r+1,i}, \psi_{pers}^{r+1,i})$ to satellite i ;
- 19 **return** $\{(\phi_{pers}^{T,i}, \psi_{pers}^{T,i})\}$ and $(\phi_{srv}^T, \psi_{srv}^T)$

to the specific environments of all satellite nodes. To address this challenge, leveraging the architecture proposed in this paper, we introduce a dual-path aggregation method wherein the server concurrently generates both a global model and a personalized model.

1) *Global Model Aggregation*: First, upon receiving the local models uploaded by individual satellites, the ground station employs the standard FedAvg algorithm to derive a global model. This model aggregates knowledge from the entire network and functions primarily as a teacher network. It is designed to transfer general knowledge to the large-scale ground station model, thereby enhancing the latter's performance. The calculation is expressed in Eq. (6):

$$\phi_r = \sum_{k \in S_r} \frac{n_k}{m_r} \phi_r^k \quad (6)$$

where k denote the index of a satellite, S_r represent the set of satellites participating in training during the r -th round, m_r be the total amount of data in the r -th round, and n_k be the amount of data held by satellite k .

2) *Attention-Based Personalized Model Aggregation*: In parallel with the generation of the global model, the server computes the cosine similarity between the models uploaded by individual satellites. These similarity metrics are subsequently utilized to derive attention scores, generating a

dedicated personalized model for each satellite client. This mechanism enables each satellite to selectively assimilate knowledge from nodes with similar data distributions while mitigating interference from nodes exhibiting significant divergence. For any two satellite clients i and j , the server calculates the cosine similarity, denoted as $Sim_{i,j}$, between their uploaded model parameters. A high degree of similarity indicates that the two satellites encounter similar traffic environments or attack patterns.

$$Sim_{i,j} = \text{Cosine}(\phi_i, \phi_j) = \frac{\phi_i \cdot \phi_j}{\|\phi_i\| \|\phi_j\|} \quad (7)$$

This process constructs an $N \times N$ similarity matrix, thereby quantifying the distribution of model similarity within the satellite constellation. Although cosine similarity captures the correlation between models, raw similarity values lack inherent normalization. Furthermore, simple linear weighting struggles to effectively differentiate between high-value models and noisy nodes; consequently, directly utilizing similarity values as aggregation weights is subject to limitations. To address this challenge, we treat the similarity values as attention scores and employ a Softmax function with a temperature coefficient to map them into a probability distribution $\alpha_{i,j}$. The calculation is formulated in Eq. (8).

$$\alpha_{i,j} = \frac{\exp(\lambda \cdot Sim_{i,j})}{\sum_{k \in S_i} \exp(\lambda \cdot Sim_{i,k})} \quad (8)$$

This approach ensures weight normalization while simultaneously enhancing aggregation selectivity via non-linear transformation. Here, λ represents the temperature coefficient, which governs the sensitivity of the aggregation process. Specifically, a larger λ encourages the model to focus on highly similar parameters, whereas a smaller λ facilitates broader knowledge fusion.

Finally, the server assembles a dedicated personalized model $\phi_{r,i}^{agg}$ for satellite i through a weighted summation as shown in Eq. (9):

$$\phi_{agg}^{r,i} = \sum_{j \in S_r} \alpha_{i,j} \cdot \phi_j^r \quad (9)$$

In contrast to the single global model ϕ_r , $\phi_{agg}^{r,i}$ is better aligned with the specific data distribution of client i , incorporating both its intrinsic data pReferences and enhanced knowledge derived from similar nodes.

3) *Knowledge Transfer and Model Distribution*: Upon the completion of aggregation, the global model ϕ_r transfers general knowledge from the satellites to the server model via the distillation mechanism detailed in the subsequent section. The updated server model then functions as a teacher, distilling its extracted deep feature knowledge into each personalized model $\phi_{agg}^{r,i}$. Finally, these personalized models are transmitted to their respective satellite clients. This dual-path aggregation mechanism ensures that the server model assimilates general knowledge from all satellite nodes, while simultaneously guaranteeing that the models distributed to the clients retain the personalization capabilities conferred by attention-based aggregation.

D. Reinforcement Learning–Assisted Multi-Granularity Bidirectional Distillation

To improve distillation efficiency and promote knowledge transfer, we propose a reinforcement learning–assisted multi-granularity bidirectional distillation method. This approach combines soft target distillation loss, feature distillation loss, and consistency distillation loss, and dynamically adjusts the weights of these three losses with the help of a reinforcement learning algorithm, thereby enhancing the efficiency of bidirectional knowledge flow.

1) *Multi-Granularity Distillation*: Soft target distillation is a distillation method based on soft logits. Soft logits refer to the outputs of the neural network’s final layer processed by the softmax function, where a temperature parameter T is applied to make the outputs smoother. In soft target distillation, the student model learns from the teacher model’s soft logits instead of relying solely on hard labels. This enables capturing rich information about class uncertainty and learning smoother decision boundaries, which helps the model better generalize to unseen data. The calculation formula for the soft label loss is as follows:

$$\begin{aligned} L_{soft} &= T^2 \cdot KL(p_t^T, p_s^T) \\ &= T^2 \cdot KL\left(\sum_{i=1}^K p_{t,i}^T \log\left(\frac{p_{t,i}^T}{p_{s,i}^T}\right)\right) \end{aligned} \quad (10)$$

where KL denotes the Kullback-Leibler divergence, which represents the information loss caused when approximating the teacher network’s soft labels p_s^T with the student network’s probability distribution p_t^T . A larger KL divergence value indicates a poorer approximation of p_t^T by p_s^T . The terms $p_{s,i}^T$ and $p_{t,i}^T$ represent the predicted probabilities of the i -th class by the student model and the teacher model, respectively.

Intermediate feature distillation forces the student model to learn certain intermediate layer features from the teacher model, either by directly matching these features or learning the transformation relationships between them. By imitating these features, the student model can achieve better generalization ability with fewer parameters. However, due to structural differences between the teacher and student models, the corresponding intermediate features often differ in dimension and number of channels. To address this issue, this paper introduces a feature adapter between the two models to transform and align the dimensions and channels of the features, thus ensuring the successful calculation of the feature distillation loss. The calculation formula is as follows:

$$L_{feat} = \sum_i \|\phi_t^{low}(b_i) - f(\phi_s^{low}(b_i))\|_2^2 \quad (11)$$

where $\phi_t^{low}(b_i)$ and $\phi_s^{low}(b_i)$ represent the intermediate features of the teacher model and student model on the input sample b_i , respectively, $f(\cdot)$ denotes the feature adapter transformation, and $\|\cdot\|_2^2$ represents the L2 norm.

Consistency distillation first performs data augmentation on input samples, and the model’s output for the original sample and the augmented sample should remain consistent [28]. Based on this principle, we apply local perturbations to the data samples and adds slight Gaussian noise. Then,

the teacher model and student model are used to predict on the original sample and the augmented sample, respectively. The teacher model’s output for the original sample is taken as the target output, and the KL divergence between the soft logits of the outputs is calculated as a consistency score for the student model’s predictions towards the teacher model. By minimizing this score, the distance between the student model and the teacher model is reduced. The calculation formula is as follows:

$$L_{cons} = T^2 \cdot KL(p_t^T, p_s'^T) \quad (12)$$

where p_t^T represents the probability distribution of the original sample predicted by the teacher model, and $p_s'^T$ represents the probability distribution of the augmented sample predicted by the student network.

By performing a weighted aggregation of the above three distillation losses, the final distillation loss is obtained. The calculation formula is as follows:

$$L_{distill} = \lambda_1 L_{soft} + \lambda_2 L_{feat} + \lambda_3 L_{cons} \quad (13)$$

where λ_1 , λ_2 , and λ_3 represent the weights corresponding to the soft target distillation loss, feature distillation loss, and consistency distillation loss, respectively.

2) *Reinforcement Learning–Assisted Bidirectional Distillation*: The three distillation methods mentioned above each have their own focus. The student model’s demand for these three distillation methods varies at different stages of training. In the early stages of training, soft target distillation can help the student model quickly learn basic knowledge, while feature distillation and consistency distillation are more helpful for further improving model performance in the later stages of training. Therefore, how to balance their weights is a key issue.

Manually adjusting the weights of the three distillation losses is a complex and time-consuming task because it involves combinations of multiple hyperparameters, and it is almost impossible for humans to try all possible combinations. To address this problem, reinforcement learning can be applied to dynamically adjust these weights. Reinforcement learning automatically explores the weight configuration space through the interaction between an agent and the environment, finding the optimal weight combination through trial and error, thereby improving efficiency.

PPO [47] is a reinforcement learning policy gradient algorithm that stabilizes policy updates by introducing a clipped objective function, avoiding drastic policy changes. Specifically, in each iteration, PPO samples a batch of data, estimates the advantage function A_t using the old policy, and optimizes the following objective function:

$$\begin{aligned} L^{CLIP}(\theta) \\ = E_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \end{aligned} \quad (14)$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ denotes the probability ratio between the new and old policies, and ρ is a hyperparameter that controls the update magnitude. This objective function improves policy performance while constraining policy changes, thereby enhancing training stability and sample efficiency. It is widely used in reinforcement learning tasks under complex environments.

In this paper, the state is represented as a vector composed of the training accuracy, training progress, and the three types of distillation losses. The action is a set of weights output by the Actor, and the reward is determined by the improvement in loss and accuracy. To ensure the stability of weights during the early stages of training, fixed weights are used initially. When the model's performance shows no improvement over several consecutive epochs during the distillation process, reinforcement learning is activated to dynamically adjust the weights. The weight adjustment process is as follows:

$$s_t = [Acc_t, Progress_t, L_{soft}, L_{feat}, L_{cons}] \quad (15)$$

$$a_t = [\lambda_1, \lambda_2, \lambda_3] \quad (16)$$

$$r_t = \Delta Acc_t - \lambda \cdot \Delta L_{total} \quad (17)$$

$$\pi_\theta(a_t|s_t) = \mathcal{N}(\mu_\theta(s_t), \sigma_\theta(s_t)^2) \quad (18)$$

$$a_t = \mu_\theta(s_t) + \sigma_\theta(s_t) \cdot \epsilon \quad (19)$$

The agent first decides, based on the current state, whether to perform random exploration or to exploit the Actor network to generate an action. If random exploration is chosen, weights are generated using a Dirichlet distribution, and a relatively small log probability is recorded. Otherwise, the action parameters are generated by the Actor network, and actions are sampled using a normal distribution. In addition, the log probability of the action is computed and stored.

After collecting data over a period of time, the Critic network is used to estimate the state-value function, and the advantage is computed using Generalized Advantage Estimation (GAE). The formula is as follows:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad (20)$$

$\delta_t = r_t + \gamma V(s_{t+1})$ is the Temporal Difference(TD) error, γ is the discount factor, λ is the GAE parameter, and $V(s_t)$ is the state-value function. Subsequently, the Actor network is updated using Equation (11). The Critic network is updated by minimizing the mean squared error between the predicted value and the actual return, in order to improve the accuracy of state-value estimation. The loss function is as follows:

$$L(\omega) = E[(V_\omega(s_t) - R_t)^2] \quad (21)$$

where $V_\omega(s_t)$ is the predicted value of the current state, and R_t is the actual return. Following the steps above, data is continuously collected during the training loop, and the reinforcement learning agent is periodically updated to dynamically optimize the loss weights and improve the model's training performance.

By applying the PPO algorithm to dynamically adjust distillation weights, the knowledge transfer from the teacher model to the student model is effectively enhanced, improving data utilization efficiency and boosting the model's generalization capability.

E. Complexity Analysis

To comprehensively evaluate the deployment feasibility of SatFedGuard in resource-constrained LEO satellite networks, we analyze the asymptotic complexity of the proposed method

Algorithm 2 RL-Assisted Multi-Granularity Bidirectional Distillation

Input : Global Model $(\phi_{global}, \psi_{global})$;
 Server Model (ϕ_{srv}, ψ_{srv}) ;
 Personalized Models $\{(\phi_{pers}^i, \psi_{pers}^i)\}$;
 Labeled dataset D^{label} ;
 Server epoch E_{srv} ;

Output: Updated Server Model $(\phi'_{srv}, \psi'_{srv})$ and
 Personalized Models $\{(\phi_{pers}^i, \psi_{pers}^i)\}$

```

1 Initialize Actor  $\theta$ , Critic  $\omega$ , Weights  $\lambda_1, \lambda_2, \lambda_3$ ;
2  $B^{label} \leftarrow$  split  $D^{label}$  into batches of size  $B$ ;
3 for direction  $dir$  in  $[Forward, Backward]$  do
4   if  $dir == Forward$  then
5     Teacher  $(\phi_T, \psi_T) \leftarrow (\phi_{global}, \psi_{global})$ ; Student
       $(\phi_S, \psi_S) \leftarrow (\phi_{srv}, \psi_{srv})$ ;
6     Target_Set  $\leftarrow \{(\phi_S, \psi_S)\}$ 
7   else
8     Teacher  $(\phi_T, \psi_T) \leftarrow (\phi_{srv}, \psi_{srv})$ ; Student_Set
       $\leftarrow \{(\phi_{pers}^i, \psi_{pers}^i)\}$ ;
9     Target_Set  $\leftarrow$  Student_Set
10  for each student model  $(\phi_S, \psi_S)$  in Target_Set do
11    for epoch  $e = 1, \dots, E_{srv}$  do
12      for batch  $b \in B^{label}$  do
13         $p^T, p^S \leftarrow \phi_T(b), \phi_S(b)$ ;
14         $L_s \leftarrow \text{CrossEntropy}(y, \hat{y})$ ;
15         $L_{soft} \leftarrow T^2 \cdot \text{KL}(p^T || p^S)$ ;
16         $L_{feat} \leftarrow \|\phi_T^{low}(b) - f(\phi_S^{low}(b))\|_2^2$ ;
17         $L_{cons} \leftarrow \text{ConsistencyLoss}(p^T, p^S)$ ;
18        if no performance improvement for
          several epochs then
19           $\pi_\theta(a_t|s_t) = \mathcal{N}(\mu_\theta(s_t), \sigma_\theta(s_t)^2)$ ;
20           $a_t = \mu_\theta(s_t) + \sigma_\theta(s_t) \cdot \epsilon$ ;
21           $\lambda_1, \lambda_2, \lambda_3 \leftarrow a_t$ ;
22           $L_{distill} \leftarrow \lambda_1 L_{soft} + \lambda_2 L_{feat} + \lambda_3 L_{cons}$ ;
23           $(\phi_S, \psi_S) \leftarrow$ 
             $(\phi_S, \psi_S) - \eta \nabla(\alpha L_s + (1 - \alpha) L_{distill})$ ;
24           $s_t = [Acc_t, Progress_t, L_{soft}, L_{feat}, L_{cons}]$ ;
25           $r_t = \Delta Acc_t - \lambda \cdot \Delta L_{total}$ ;
26          Update  $\theta, \omega$  via PPO;
27        if  $dir == Forward$  then
28           $(\phi_{srv}, \psi_{srv}) \leftarrow \text{EMA}(\phi_S, \text{EMA}(\psi_S))$ ;
29        else
30           $\{(\phi_{pers}^i, \psi_{pers}^i)\} \leftarrow \{(\text{EMA}(\phi_S^i), \text{EMA}(\psi_S^i))\}$ 
31 return  $(\phi'_{srv}, \psi'_{srv})$  and  $\{(\phi_{pers}^i, \psi_{pers}^i)\}$ 

```

during both training and communication phases. We decompose the overall overhead into atomic operations for satellite nodes and the ground server, respectively.

1) *Computational Complexity:* For satellite nodes, the computational burden primarily stems from the iterative local training process, which involves the forward and backward propagation of the backbone network and the calculation of the contrastive loss. Let the lightweight satellite model consist of L_{conv} convolutional layers and L_{fc} fully connected layers. For the l -th convolutional layer, we define the spatial size of the feature map as $M_l (H_l \times W_l)$, the number of input and

output channels as C_{l-1} and C_l , and the kernel size as K_l . For the j -th fully connected layer, the number of input and output neurons are U_{j-1} and U_j . We first define the computational complexity of the forward propagation for a single batch of size B as F_{small} :

$$F_{small} = O \left(B \left[\sum_{l=1}^{L_{conv}} M_l C_{l-1} C_l K_l^2 + \sum_{j=1}^{L_{fc}} U_{j-1} U_j \right] \right) \quad (22)$$

Considering that the backward propagation for gradient calculation typically incurs approximately twice the computational cost of the forward pass, the cost for one training step is $3F_{small}$. Additionally, the Barlow Twins algorithm requires computing the cross-correlation matrix, introducing an extra cost $O(\Theta_{loss}) = O(B \cdot D_{emb}^2)$. Combining these components, in each global communication round, the satellite performs E_s epochs of training on the local dataset of size $|D_k|$. Therefore, the total training computational complexity for a single satellite, O_{sat} , is derived as:

$$O_{sat} = O \left(T \cdot E_s \cdot \frac{|D_k|}{B} \cdot (3F_{small} + \Theta_{loss}) \right) \quad (23)$$

Since a shallow architecture with $L_{conv} = 3$ is adopted and D_{emb} is restricted, F_{small} and Θ_{loss} are maintained at low levels, ensuring the efficiency of onboard training.

For the ground server, the computational cost comprises the attention-based dual-path aggregation, reinforcement learning decision-making, and fine-grained bidirectional distillation training. First, the server executes the dual-path aggregation. This involves calculating the cosine similarity matrix for N satellite models and generating N distinct personalized models via attention-weighted summation. The complexity for calculating the similarity matrix and aggregating personalized parameters is $O(N^2 \cdot |\theta_{sat}|)$. Second, the PPO agent update incurs a complexity of $O(\Theta_{rl})$. Third, for the distillation phase, we model the ground-side DenseNet which consists of N_{blk} dense blocks and N_{tr} transition layers. Let the b -th dense block have L_b layers with a growth rate k . For the i -th layer in this block, let $C_{b,i}^{in}$ be the input channels and M_b be the feature map size. Assuming a bottleneck structure, we first define the computational cost for the b -th dense block including the transition layer as $O_{block}^{(b)}$. This consists of L_b bottleneck layers and one transition layer:

$$O_{block}^{(b)} = M_b \left[\sum_{i=1}^{L_b} (4k \cdot C_{b,i}^{in} + 36k^2) + \theta(C_{b,i}^{out})^2 \right] \quad (24)$$

where $36k^2$ accounts for the 3×3 convolution cost. Consequently, the total forward propagation complexity F_{dense} is the summation over all blocks:

$$F_{dense} = O \left(B \sum_{b=1}^{N_{blk}} O_{block}^{(b)} \right) \quad (25)$$

The bidirectional distillation mechanism unfolds in two distinct stages. First, the forward distillation employs the small model as an inference teacher with a cost of F_{small} to guide the training of the DenseNet student which incurs a cost of $3F_{dense}$. Conversely, the backward distillation reverses these

roles, requiring the DenseNet to perform inference with a cost of F_{dense} while the small model undergoes training with a cost of $3F_{small}$. By aggregating these operations, the total computational burden per batch amounts to $4(F_{small} + F_{dense})$. For notational brevity, we define this composite distillation cost as F_{dist} . Consequently, the total computational complexity for the ground server, O_{srv} , is formulated as:

$$O_{srv} = O \left(T \left(N^2 |\theta_{sat}| + \Theta_{rl} + E_g \frac{|D_g|}{B} F_{dist} \right) \right) \quad (26)$$

This formulation highlights that while the ground station handles the heavy computation of F_{dense} and the $O(N^2)$ attention mechanism, it significantly relieves the computational burden on satellite nodes.

2) *Communication Complexity*: In terms of communication overhead, the bandwidth consumption depends on the model parameter size $|\theta_{sat}|$, calculated as $|\theta_{sat}| \approx \sum_{l=1}^{L_{conv}} C_{l-1} C_l K_l^2 + \sum_{j=1}^{L_{fc}} U_{j-1} U_j$. In each round, a satellite uploads the updated local parameters, leading to an uplink complexity of $O(N \cdot T \cdot |\theta_{sat}|)$. For the downlink, unlike traditional Federated Learning which broadcasts a single global model, the server transmits a unique personalized model $\phi_{agg}^{r,i}$ to each corresponding satellite i based on the dual-path aggregation. This results in a downlink complexity of $O(N \cdot T \cdot |\theta_{sat}|)$. The total communication complexity is $O(2 \cdot N \cdot T \cdot |\theta_{sat}|)$. Compared to transmitting raw traffic data where complexity scales linearly with data volume $|D_k|$, SatFedGuard decouples communication costs from data scale, offering significant bandwidth advantages for large-scale satellite networks.

V. EVALUATIONS AND DISCUSSION

A. Dataset Description and Preprocessing

To evaluate the effectiveness of the proposed method, two open-source intrusion detection datasets are used: the STIN dataset and the STI dataset. These datasets were preprocessed to meet the experimental requirements. To align with the design objectives of SatFedGuard, we explicitly restricted the datasets to satellite-domain traffic. Given that the method is specifically tailored for resource-constrained on-board environments, we concentrated on detecting threats within the space segment. This approach enables the model to capture domain-specific features more effectively, thereby maximizing detection efficacy within its intended operational scenarios.

The STIN dataset [17], released in 2020, is a space-ground integrated intrusion detection dataset, which includes the SAT20 (satellite network traffic) and TER20 (terrestrial network traffic) datasets. It was collected from a large-scale Linux-based prototype system, where the ground network uses the TCP/IP protocol and the satellite network uses the DTN protocol. SAT20 contains two types of attack traffic: SYN_DDoS and UDP_DDoS. TER20 includes a total of seven types of attack traffic targeting terrestrial networks, such as backdoor attacks, botnets, web attacks, and four types of DDoS attacks. In this work, the SAT20 dataset is selected as the experimental dataset, consisting of 31 features and 82,320 network traffic records.

The STI dataset [34], released in 2024, is a space-ground integrated intrusion detection dataset containing eight labels:

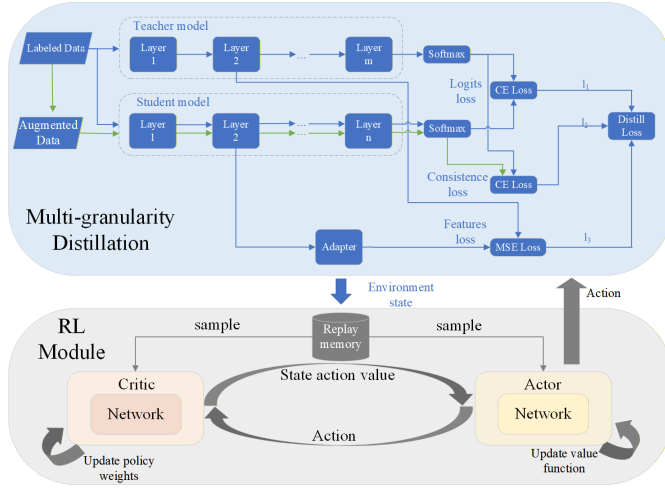


Fig. 3. Reinforcement learning-assisted multi-granularity distillation diagram.

benign, signal jamming, UDP flooding, interference, DoS, DDoS, brute force, and infiltration. Among them, signal jamming, UDP flooding, and interference originate from the satellite domain, while DoS, DDoS, brute force, and infiltration come from the ground domain. In this study, only benign traffic and attack traffic samples from the satellite domain are selected, resulting in a dataset with 20 features and 1,221,557 records.

The datasets are split into training and testing sets with a ratio of 8:2. To improve training efficiency, the traffic data is first preprocessed to identify and handle outliers, missing values, or incorrect values. Then, the most relevant and effective features are selected to reduce data dimensionality. The data is scaled to a specific range to prevent large disparities between feature values, thereby stabilizing the model, accelerating convergence, and enhancing generalization ability. Finally, since normal samples far outnumber abnormal ones in anomaly detection tasks, a weighted average sampling algorithm is used to balance the distribution of positive and negative samples, addressing the issue of dataset imbalance. The preprocessing process includes five steps: data cleaning, feature selection, type conversion, data normalization, and weighted random sampling.

Due to differences in satellite orbits and overlapping geographic coverage, the traffic data collected by individual nodes exhibits both heterogeneity and redundancy. To comprehensively evaluate model performance, we adopt a data partitioning strategy based on the Dirichlet Distribution, constructing two comparative experimental scenarios by adjusting the heterogeneity parameter α . For the Non-IID scenario, we set $\alpha = 0.5$ to simulate the real-world phenomenon of class imbalance in traffic, where the sample distribution across nodes varies significantly. Conversely, for the IID scenario, we set $\alpha = 100$ to simulate an ideal uniform distribution, where sample distributions across nodes tend to be consistent.

In the specific implementation, the original dataset is partitioned into labeled and unlabeled subsets, with the server retaining a fixed quantity of labeled data, denoted as $N(D^l)$. For the remaining unlabeled data, we employ a sampling

mechanism with replacement to allocate subsets to each satellite, setting the data overlap ratio at $\gamma = 0.1$ to align with the characteristics of overlapping geographic coverage inherent in constellation operations. Under this configuration, the specific sample distributions of the STIN and STI datasets across nodes are illustrated in Fig. 4.

B. Experiment Setup

In this study, all experiments were conducted on the Ubuntu 22.04 LTS platform, equipped with an NVIDIA GeForce RTX 4070 Ti. The larger model at the ground station is DenseNet, and the satellite-side model consists of three convolutional layers and two fully connected layers. These models were implemented using PyTorch 1.13.1 and Python 3.10.8. To adapt to the two datasets, the model architecture was appropriately adjusted. In the experiments, the number of labeled data samples and the number of unlabeled samples stored on each satellite were fixed. As the number of satellites changes, the total number of unlabeled samples participating in training changes linearly.

We use accuracy, precision, recall, and F1-score as evaluation metrics to assess the performance of the model. The calculation method of these metrics is as follows:

Accuracy: The proportion of all correctly predicted traffic samples to the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (27)$$

Precision: The proportion of correctly predicted normal traffic samples among all samples predicted as normal traffic.

$$Precision = \frac{TP}{TP + FP} \quad (28)$$

Recall: The proportion of correctly predicted normal traffic samples among all actual normal traffic samples.

$$Recall = \frac{TP}{TP + FN} \quad (29)$$

Macro F1-score: Considering the quantity imbalance among different traffic categories, we choose to calculate the macro F1-score.

$$Macro\ F1\ score = \frac{1}{K} \sum_{i=1}^K 2 \cdot \frac{Precision_i \cdot Recall_i}{Precision_i + Recall_i} \quad (30)$$

where K is the number of classes, and the macro F1-score measures the average of the F1 scores calculated separately for each class.

C. Performance Under Different Factors

In this section, we evaluate the performance of the proposed method across both IID and Non-IID scenarios, investigating the impact of various factors including the number of participating satellites, the number of local training epochs, and the total number of global training rounds. Regarding the training configuration, the number of local epochs is set to 1, 5, and 10, respectively, while the maximum number of global communication rounds is fixed at 50. To ensure statistical reliability, each experiment was repeated five times.

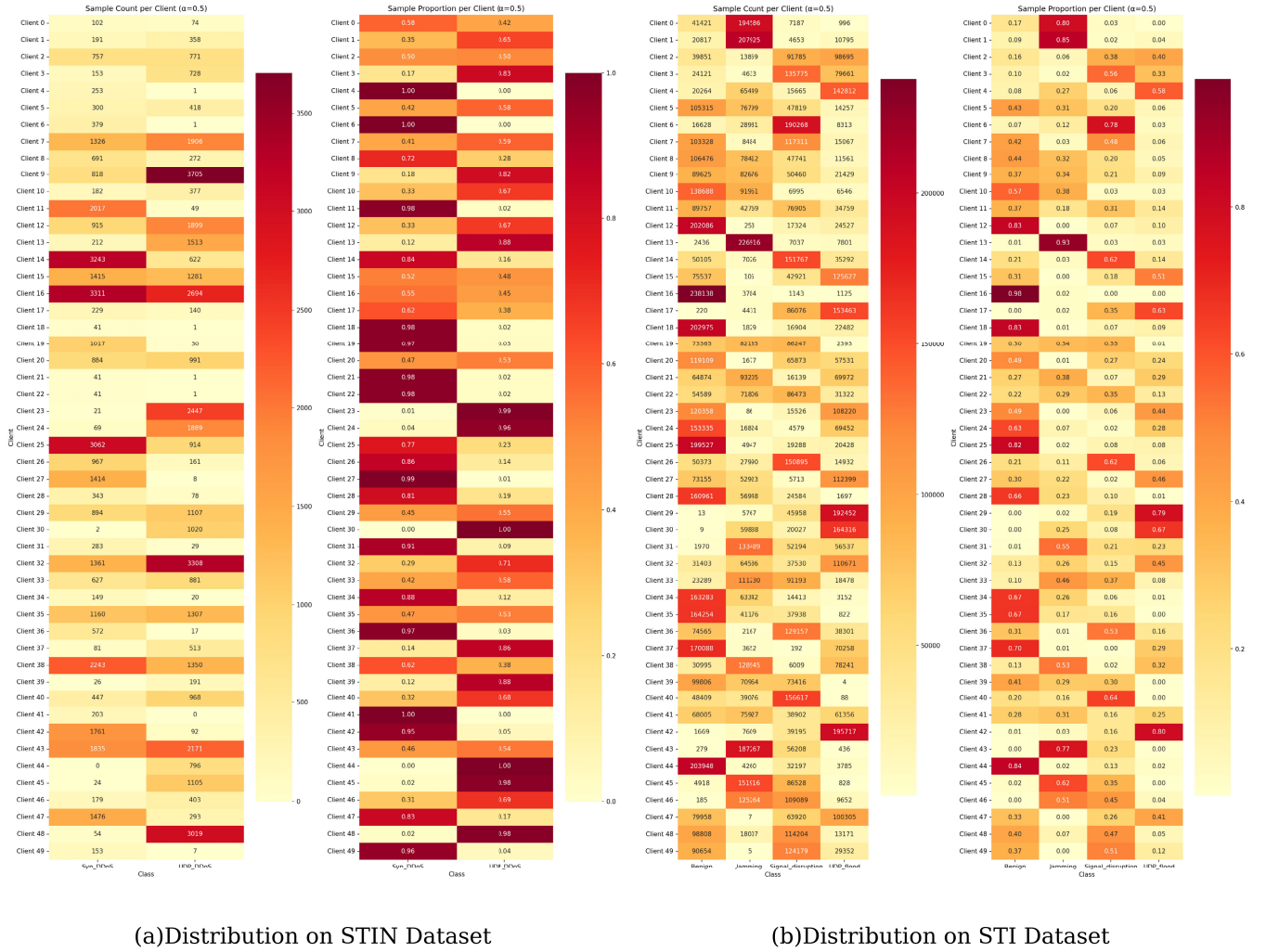


Fig. 4. Visualization of Non-IID data based on dirichlet partitioning with $\alpha = 0.5$. (a) STIN dataset; (b) STI dataset.

The specific impacts of these factors on model performance metrics under IID and Non-IID scenarios are illustrated in Fig. 5 and Fig. 6, respectively.

Based on the model's performance on the two datasets, we observe that as the number of communication rounds increases, the model's accuracy continuously improves until it eventually converges. Increasing the number of satellite clients participating in training significantly boosts the model's accuracy.

For the IID scenario, experimental results on the STIN dataset demonstrate that with the number of satellite clients N_c set to 10, 30, and 50, the final model accuracies converge to 92.9%, 94.0%, and 94.4%, respectively. Notably, the accuracy at $N_c = 50$ reflects an improvement of 0.4% and 1.5% over $N_c = 30$ and $N_c = 10$. Similarly, on the STI dataset, the accuracies converge to 96.4%, 98.4%, and 99.8%, respectively, indicating performance gains of 1.4% and 3.4% at $N_c = 50$ compared to the cases of $N_c = 30$ and $N_c = 10$. In the Non-IID scenario, the proposed method maintains robust performance. On the STIN dataset, for $N_c \in \{10, 30, 50\}$, the final accuracies reach 91.2%, 91.8%, and 92.7%, respectively. The performance at $N_c = 50$ exhibits increases of 0.9% and

1.5% relative to $N_c = 30$ and $N_c = 10$. Furthermore, on the STI dataset, the model converges to accuracies of 96.1%, 97.2%, and 97.8%, respectively, with the configuration of $N_c = 50$ achieving improvements of 0.6% and 1.7% over the $N_c = 30$ and $N_c = 10$ settings. This improvement mainly benefits from the increased number of available training samples, enabling the model to learn from a broader range of samples and extract more comprehensive and representative features, effectively enhancing the generalization ability of the global model and thereby improving accuracy.

It is noteworthy that distinct behavioral patterns emerge between IID and Non-IID scenarios as the number of local training epochs (E) increases. In the IID scenario, increasing local epochs consistently accelerates model convergence. Specifically, on the STIN dataset, for $E = 1, 5, 10$, the model requires 35, 31, and 28 rounds, respectively, to achieve convergence. Similarly, on the STI dataset, the required rounds are reduced to 30, 20, and 15, respectively. This improvement is attributed to the uniform data distribution across nodes under the IID assumption. Consequently, performing multiple local updates enables clients to better exploit local data characteristics; the model updates transmitted to the ground server are

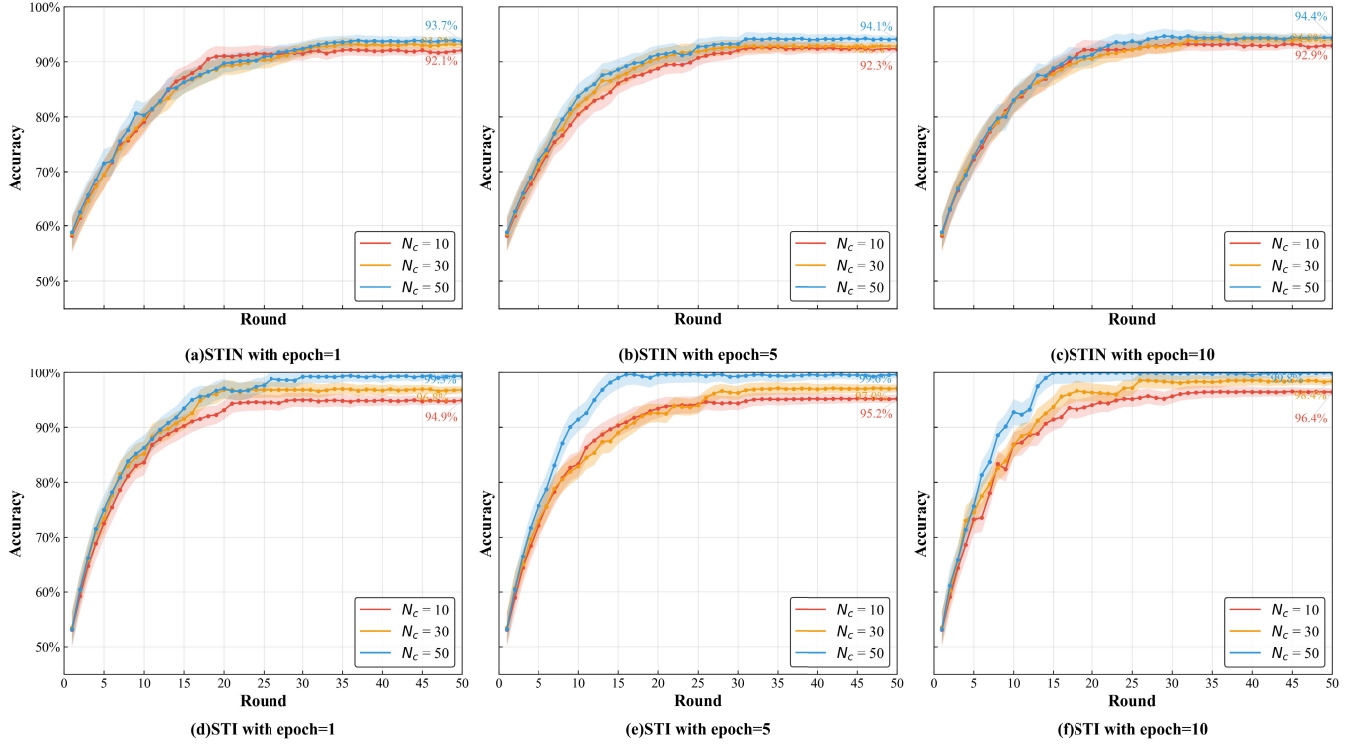


Fig. 5. Performance evaluation on IID data.

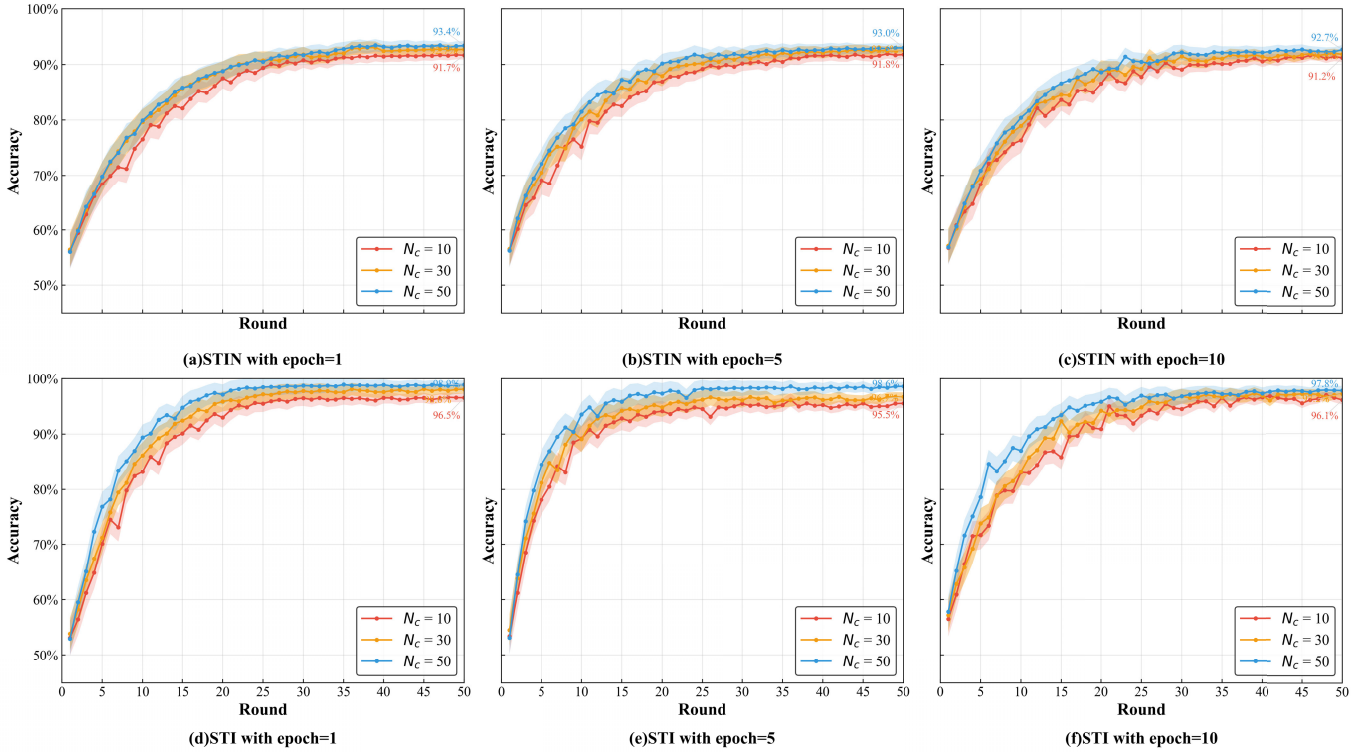


Fig. 6. Performance evaluation on Non-IID data.

thus more refined and representative, effectively capturing the global data distribution with fewer communication rounds.

Conversely, in the Non-IID scenario, an increase in E does not guarantee a reduction in communication rounds; rather, it

may induce fluctuations or even lead to an increase. For the STIN dataset, with $E = 1, 5, 10$, the convergence rounds are recorded as 37, 29, and 31, respectively. On the STI dataset, the rounds are 28, 25, and 30, respectively. This phenomenon

TABLE II

OVERALL TESTING ACCURACY, PRECISION, RECALL AND F1 SCORE WITH COMPARED STATE-OF-THE-ART METHODS UNDER IID SCENARIO (%)

Algorithm	STIN				STI			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
SAT-IDS	93.33 $\pm$ 0.79	93.33 $\pm$ 1.08	94.12 $\pm$ 0.48	93.30 $\pm$ 0.52	95.00 $\pm$ 1.18	97.73 $\pm$ 0.93	91.67 $\pm$ 1.27	93.81 $\pm$ 0.81
CANSat-IDS	91.85 $\pm$ 0.45	92.84 $\pm$ 0.41	92.01 $\pm$ 0.36	91.82 $\pm$ 0.29	96.11 $\pm$ 1.11	96.18 $\pm$ 1.08	94.79 $\pm$ 0.49	95.43 $\pm$ 0.53
STINIDF	94.55 $\pm$ 0.24	94.99 $\pm$ 0.11	94.66 $\pm$ 0.46	94.54 $\pm$ 0.20	99.81 $\pm$ 0.17	99.77 $\pm$ 0.11	99.80 $\pm$ 0.17	99.78 $\pm$ 0.12
SVM	87.54 $\pm$ 0.27	89.86 $\pm$ 0.33	87.79 $\pm$ 0.26	87.41 $\pm$ 0.31	95.41 $\pm$ 2.05	96.11 $\pm$ 1.62	93.36 $\pm$ 3.83	94.65 $\pm$ 1.94
KNN	93.08 $\pm$ 0.79	93.46 $\pm$ 1.08	93.19 $\pm$ 0.48	93.08 $\pm$ 0.52	91.91 $\pm$ 1.18	94.86 $\pm$ 0.93	85.13 $\pm$ 1.27	88.45 $\pm$ 0.81
Naive Bayes	87.56 $\pm$ 0.54	89.87 $\pm$ 0.59	87.81 $\pm$ 0.45	87.43 $\pm$ 0.32	85.87 $\pm$ 2.19	80.73 $\pm$ 0.68	86.24 $\pm$ 1.15	82.71 $\pm$ 1.39
MLP	88.01 $\pm$ 0.23	90.14 $\pm$ 0.14	88.25 $\pm$ 0.47	87.89 $\pm$ 0.19	97.87 $\pm$ 1.18	96.87 $\pm$ 2.23	98.01 $\pm$ 1.63	97.42 $\pm$ 1.08
Method of [41]	88.01 $\pm$ 0.82	90.17 $\pm$ 0.34	88.25 $\pm$ 0.27	87.89 $\pm$ 0.16	97.89 $\pm$ 1.18	98.97 $\pm$ 0.92	95.78 $\pm$ 1.28	97.22 $\pm$ 1.06
Method of [34]	85.53 $\pm$ 1.47	86.69 $\pm$ 0.49	85.71 $\pm$ 2.31	85.45 $\pm$ 1.17	79.12 $\pm$ 6.59	77.13 $\pm$ 7.47	65.47 $\pm$ 8.42	66.79 $\pm$ 8.05
SatFedGuard	94.38 $\pm$ 0.21	94.62 $\pm$ 0.30	94.31 $\pm$ 0.32	94.38 $\pm$ 0.21	99.80 $\pm$ 0.11	99.78 $\pm$ 0.12	99.82 $\pm$ 0.06	99.80 $\pm$ 0.11

TABLE III

OVERALL TESTING ACCURACY, PRECISION, RECALL AND F1 SCORE UNDER NON-IID SCENARIO (%)

Algorithm	STIN				STI			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
SAT-IDS	90.81 $\pm$ 0.82	90.78 $\pm$ 1.13	91.52 $\pm$ 0.58	90.80 $\pm$ 0.58	92.56 $\pm$ 1.23	95.11 $\pm$ 1.01	89.17 $\pm$ 1.33	91.39 $\pm$ 0.88
CANSat-IDS	89.24 $\pm$ 0.49	90.28 $\pm$ 0.48	89.43 $\pm$ 0.42	89.30 $\pm$ 0.37	93.61 $\pm$ 1.18	93.72 $\pm$ 1.16	92.23 $\pm$ 0.48	92.96 $\pm$ 0.60
STINIDF	91.52 $\pm$ 0.39	91.97 $\pm$ 0.22	91.60 $\pm$ 0.55	91.48 $\pm$ 0.28	96.75 $\pm$ 0.28	96.68 $\pm$ 0.18	96.72 $\pm$ 0.27	96.71 $\pm$ 0.19
SVM	85.04 $\pm$ 0.35	87.24 $\pm$ 0.42	85.11 $\pm$ 0.39	84.88 $\pm$ 0.38	92.96 $\pm$ 2.17	93.54 $\pm$ 1.72	90.85 $\pm$ 3.91	92.10 $\pm$ 2.04
KNN	90.81 $\pm$ 0.85	90.92 $\pm$ 1.19	90.68 $\pm$ 0.36	90.30 $\pm$ 0.48	89.48 $\pm$ 1.25	92.39 $\pm$ 1.05	82.04 $\pm$ 1.35	85.96 $\pm$ 0.88
Naive Bayes	84.26 $\pm$ 0.62	86.22 $\pm$ 0.61	84.23 $\pm$ 0.42	83.95 $\pm$ 0.40	82.37 $\pm$ 2.28	77.14 $\pm$ 0.79	82.63 $\pm$ 1.21	79.17 $\pm$ 1.47
MLP	82.34 $\pm$ 0.32	84.61 $\pm$ 0.22	82.73 $\pm$ 0.73	82.39 $\pm$ 0.28	91.31 $\pm$ 1.25	90.39 $\pm$ 2.35	91.54 $\pm$ 1.57	90.88 $\pm$ 1.10
Method of [41]	81.96 $\pm$ 1.12	84.07 $\pm$ 0.62	82.14 $\pm$ 0.46	81.75 $\pm$ 0.41	90.87 $\pm$ 1.66	91.72 $\pm$ 1.19	89.67 $\pm$ 1.69	90.16 $\pm$ 1.49
Method of [34]	81.45 $\pm$ 1.85	82.33 $\pm$ 0.87	81.64 $\pm$ 2.64	81.38 $\pm$ 1.85	77.09 $\pm$ 6.87	75.13 $\pm$ 7.72	63.31 $\pm$ 8.71	65.75 $\pm$ 8.88
SatFedGuard	93.40 $\pm$ 0.35	93.65 $\pm$ 0.41	93.30 $\pm$ 0.38	93.38 $\pm$ 0.32	98.88 $\pm$ 0.22	98.85 $\pm$ 0.19	98.92 $\pm$ 0.15	98.88 $\pm$ 0.18

arises from the high heterogeneity of data distributions across individual satellites. Excessive local iterations cause the local model to overfit its specific local distribution, resulting in significant weight divergence from the global model. As a result, aggregating these divergent models becomes more challenging, thereby impeding the global convergence process.

Furthermore, a similar contrast is observed regarding final accuracy. In the IID scenario, increasing E typically enhances the final accuracy. However, in the Non-IID scenario, a higher E may conversely lead to accuracy degradation. This degradation is attributed to the misalignment between local optimization objectives and the global objective caused by data heterogeneity; specifically, over-optimized local models experience client drift, deviating from the global optimum.

D. Comparison Experiments

1) *Comparison With Supervised Learning Models:* To validate the effectiveness of the proposed method, we conducted comparative experiments against several supervised learning benchmarks in both IID and Non-IID scenarios across two datasets. The baseline methods include SAT-IDS [31], CANSat-IDS [32], STINIDF [34], SVM [48], KNN [49], Naive Bayes [50], and MLP [51]. All baseline methods were trained on labeled datasets. To ensure statistical reliability,

each experiment was repeated five times. We report the mean values and standard deviations for Accuracy, Precision, Recall, and F1-score. The results are presented in Table II and Table III, respectively.

In the IID setting, the proposed method generally outperforms the supervised learning baselines across both datasets. Specifically, on the STIN dataset, although our method trails the state-of-the-art STINIDF by a marginal 0.16%, it establishes a significant advantage over the next-best baseline, achieving improvements of 1.05%, 1.29%, 0.19%, and 1.08% in Accuracy, Precision, Recall, and F1-score, respectively. On the STI dataset, our method achieves performance comparable to STINIDF and consistently surpasses all other baselines; compared to the runner-up method, it yields increases of 1.93%, 2.91%, 1.81%, and 2.38% in Accuracy, Precision, Recall, and F1-score, respectively. This is because in scenarios with scarce labeled data but abundant unlabeled data, our method can fully exploit the distribution structure of unlabeled data to help the model learn more comprehensive features, thereby improving generalization. STINIDF uses a differentially private conditional diffusion model (DP-CDM) to generate synthetic data that conforms to STIN traffic generation constraints, partially mitigating the impact of limited labeled data, thus achieving nearly the same results as our method.

In the Non-IID setting, data heterogeneity poses significant challenges, leading to a general performance decline across all models; however, the proposed method demonstrates remarkable resilience, maintaining high performance levels. On the STIN dataset, compared to the runner-up method, our approach achieves improvements of 1.88% in Accuracy, 1.68% in Precision, 1.7% in Recall, and 1.9% in F1-score. Similarly, on the STI dataset, it outperforms the next-best method with gains of 2.13%, 2.13%, 2.2%, and 2.17% in Accuracy, Precision, Recall, and F1-score, respectively. This robustness stems from two key factors. First, our method effectively leverages the distributional structure of massive unlabeled data alongside the deep features of limited labeled data, enabling the model to learn more comprehensive representations even when local data exhibits Non-IID distributions. Second, the attention-based personalized aggregation mechanism employed in this paper effectively mitigates the performance degradation caused by divergences in data distribution.

2) *Comparison With Sota*: In our experiments, we also compared with the current state-of-the-art semi-supervised intrusion detection methods, including two federated semi-supervised learning approaches. The method proposed in [40] is the same as ours, where clients store unlabeled data and the server stores labeled data; the method proposed in [33] assumes both labeled and unlabeled data are stored on clients, with the server only aggregating models.

Table II and table III presents our experimental results. Our method significantly outperforms both approaches. Compared to the method proposed in [33], under the IID scenario, our method improves the F1-score by 8.93% and 33.01% on the STIN and STI datasets, respectively. It is noteworthy that this performance advantage is significantly amplified in the Non-IID scenario, reaching 12% and 33.13%, respectively. Beyond the robustness conferred by our personalized aggregation method tailored for Non-IID scenarios, this improvement is attributed to introducing a larger model on the server side and employing reinforcement learning-assisted multi-granularity distillation. The larger model has stronger feature extraction capabilities than smaller models, capturing deeper data characteristics, while reinforcement learning-assisted multi-granularity distillation facilitates efficient knowledge transfer between models, boosting performance.

In comparison with [40], under the IID scenario, our method improves the F1-score by 6.49% and 2.58% on the STIN and STI datasets, respectively. This performance advantage is further amplified in the Non-IID scenario, expanding to 11.63% and 8.72%, respectively. Although method proposed in [40] alleviates reliance on labeled data to some extent, it struggles to effectively guide model optimization in scenarios with extremely scarce labeled data on clients. Our method can effectively capture intrinsic data features in a fully unlabeled client setting, then optimize the model using a more powerful large model and a small amount of labeled data. Thus, our method shows clear performance superiority over it, especially on the larger-scale STI dataset.

E. Ablation Study

To validate the effectiveness of the proposed method, we conducted ablation studies, including: (1) performing supervised learning directly on local data after global model aggregation without utilizing larger model for bidirectional distillation; (2) using only basic soft-label distillation instead of the multi-grained distillation scheme during distillation; (3) employing the multi-grained distillation scheme during distillation, but without dynamically adjusting weights via reinforcement learning; (4) substituting the proposed personalized aggregation algorithm with the standard FedAvg algorithm; and (5) investigating the impact of the contrastive learning module on feature representations via visualization techniques. The results corresponding to these three ablation experiments are shown in Figure 7.

1) *Impact of Bidirectional Distillation*: To verify the effectiveness of introducing larger model for bidirectional distillation, we compared our method with one that does not use larger model for bidirectional distillation. Specifically, after global model aggregation, a classifier was attached to the global model, and then supervised learning was performed based on server-side labeled data. The model trained by this method was then compared with our proposed model. Experimental results demonstrate that under the IID scenario, the proposed method achieves F1-score improvements of 2.68% and 21.56% on the STIN and STI datasets, respectively. Furthermore, in the Non-IID scenario, F1-score improvements of 3.58% and 22.57% are observed on the STIN and STI datasets, respectively. This indicates that introducing larger model for bidirectional distillation can effectively improve the model's classification performance and enhance its generalization ability. This is because larger model possess deeper network structures, enabling them to capture more profound data features. Furthermore, due to their greater number of network parameters, larger model exhibit stronger generalization capabilities, performing more excellently when faced with varying data distributions. By transferring knowledge from larger model to the smaller satellite-shared models through distillation, the performance on the satellite-side has been further boosted.

2) *Impact of Multi-Grained Distillation*: To validate the effectiveness of multi-grained bidirectional distillation, we replaced it with classic soft-label distillation for comparison. The results show that in the IID scenario, F1-score improvements of 3.22% and 1.47% are achieved on the STIN and STI datasets, respectively; meanwhile, in the Non-IID scenario, the F1-scores on the STIN and STI datasets improve by 3.12% and 1.46%, respectively. This demonstrates that the proposed multi-grained distillation can effectively enhance distillation efficiency and improve the detection performance of small edge models. This is because simple soft-label distillation transmits relatively singular information, whereas multi-grained distillation can capture information from multiple levels, transmitting the teacher model's knowledge more comprehensively and thus increasing the training efficiency of the student model.

3) *Impact of Reinforcement Learning*: To verify the effectiveness of introducing reinforcement learning for dynamically

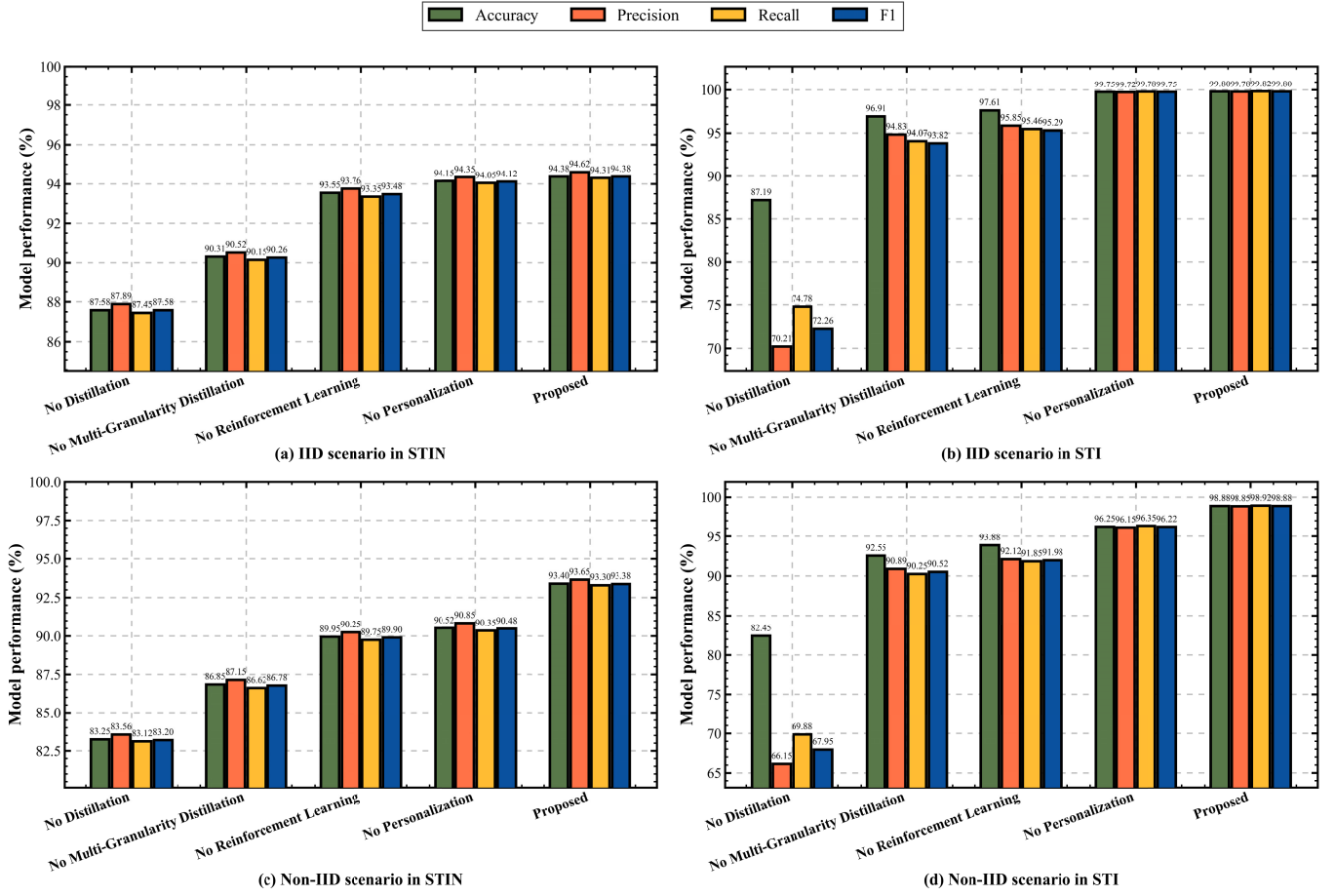


Fig. 7. Ablation study results on the STIN and STI dataset.

adjusting distillation weights, we conducted experiments where only fixed weights were used during the distillation process, without dynamic adjustment via reinforcement learning. Experimental results show that in the IID scenario, F1-score improvements of 0.64% and 4.46% are achieved on the STIN and STI datasets, respectively. In the Non-IID scenario, the F1-scores on the STIN and STI datasets improve by 0.58% and 4.24%, respectively. This indicates that dynamically adjusting weights with reinforcement learning can effectively improve distillation efficiency, thereby promoting the enhancement of global model performance. The contributions of different distillation branches to the student model are dynamic, and fixed weights cannot perceive these dynamic demands, potentially over-relying on or neglecting certain distillation signals, ultimately leading to suboptimal knowledge transfer.

4) *Impact of Personalized Aggregation:* To validate the effectiveness of the attention-based dual-path aggregation method, we employed a baseline approach utilizing the standard FedAvg algorithm to aggregate a single global model. In this setup, after the global model transfers general knowledge to the ground station model via distillation, the ground station model functions as a teacher, distilling knowledge back into the global model, which is subsequently distributed to individual satellites. Experimental results indicate that in the IID scenario, the F1-scores on the STIN and STI datasets improved

by only 0.26% and 0.05%, respectively. Conversely, in the Non-IID scenario, the improvements on these datasets were significantly more pronounced, reaching 2.9% and 2.66%, respectively. This disparity is attributed to the inherent characteristics of the data distributions. In the IID scenario, where the data distribution across satellite nodes tends to be uniform, the general global model is already capable of fitting the data effectively; consequently, the marginal gain yielded by the personalized aggregation algorithm is limited. In contrast, in the Non-IID scenario, data heterogeneity induces significant divergence between the local optima of individual nodes and the global model. The proposed personalized aggregation mechanism addresses this by enabling each satellite node to retain unique local feature information while maintaining collaboration. This effectively corrects the aforementioned divergence, thereby unlocking substantial potential for performance enhancement within heterogeneous data environments.

5) *Impact of Contrastive Learning via Visualization:* To visually evaluate the feature representation capability of the proposed method and investigate the quality of features learned by the contrastive learning module, we employed the t-Distributed Stochastic Neighbor Embedding (t-SNE) technique for dimensionality reduction and visualization analysis. Specifically, we projected high-dimensional data into a two-dimensional space to compare two critical stages within the

TABLE IV
RESOURCE CONSUMPTION AND EFFICIENCY ANALYSIS ON STI DATASET

Method	Params (K)	FLOPs (K)	Train Lat. (ms)	Infer Lat. (ms)	Infer Energy (μ J)	Train Energy (mJ)
Ours	29.518	51.597	8040.33	0.2039	19.805	780.96
SAT-IDS	41.508	42.573	9531.71	0.2368	20.136	810.48
Method of [34]	73.608	332.480	19331.33	0.6316	116.513	3566.05
STINIDF	18.824	19.095	6343.37	0.1557	9.059	369.06
Method of [41]	1.841	1.740	4611.64	0.0768	0.837	50.27

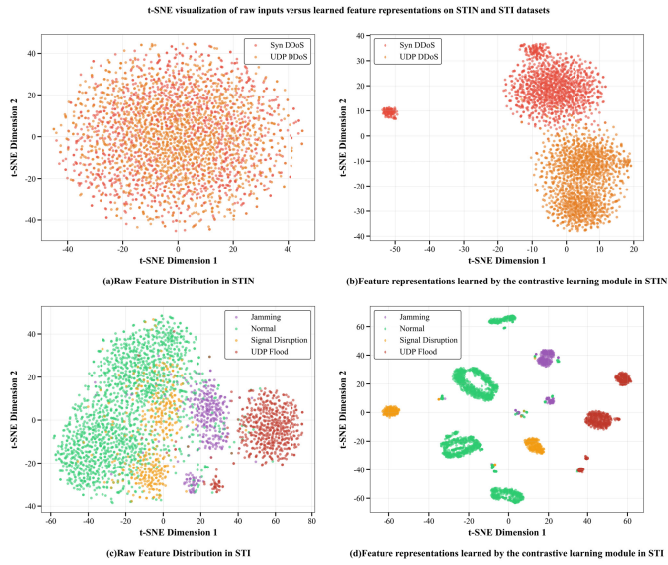


Fig. 8. t-SNE visualization comparison between raw inputs and contrastive learning learned feature representations on STIN and STI datasets.

processing pipeline: the raw input features and the latent representations extracted by the penultimate layer of the contrastive learning module. This comparative experiment was conducted on both the STIN and STI datasets. By randomly sampling instances from the test sets, we were able to visually assess the impact of the model on data distribution via feature space transformation.

The visualization results presented in Fig. 8 reveal a distinct transition in feature distribution, evolving from class entanglement in the raw space to structured clustering within the latent representations. As depicted in the left column, the raw features of different traffic types exhibit severe overlap with indistinct boundaries, posing significant challenges for linear classification. In stark contrast, as shown in the right column, the contrastive learning module successfully extracts discriminative representations characterized by high intra-class compactness and inter-class separability. This process maps the originally non-linearly separable data into a feature space defined by clear decision boundaries. The resulting high-quality representations significantly reduce the complexity for downstream classifiers. This visualization demonstrates the module's capability to effectively extract distinct semantic

representations, reconstructing the non-linearly entangled raw features into a linearly separable space.

F. Efficiency Analysis

1) *Computation Overhead*: Table IV presents a comprehensive evaluation of the model's computational overhead and resource efficiency. The results indicate that the proposed method achieves an optimal trade-off between detection performance and resource consumption, which is critical for resource-constrained LEO satellite networks.

From the perspective of computational burden, our method significantly alleviates the load on the satellite payload compared to complex baselines such as [33] and SAT-IDS. Specifically, compared to [33], our method reduces FLOPs by 84.5% and energy consumption by approximately 83%. This substantial reduction is attributed to our streamlined on-board model design, which eliminates the redundant feature extraction layers found in [33], thereby effectively extending on-board battery life while minimizing processing latency. Furthermore, we note that while [40] and STINIDF exhibit lower parameter counts and latency, efficiency must be evaluated in conjunction with effectiveness. Although [40] is extremely lightweight, its overly simplistic structure fails to capture complex attack patterns, resulting in detection accuracy inferior to that of our method. Compared to STINIDF, our method increases the parameter count by approximately 10K and inference latency by roughly 0.048 ms. This slight increase in overhead represents a necessary computational cost incurred to enhance detection accuracy. Unlike STINIDF, which may underperform under severe data heterogeneity, our method effectively models local distribution features. We argue that sacrificing negligible computational resources in exchange for significant F1-score improvements in Non-IID scenarios represents a highly cost-effective trade-off for building resilient satellite security systems.

From the perspective of practical deployment, the overhead of the proposed method is well within the capabilities of modern satellite processors. In terms of storage, a parameter count of approximately 29.5K implies a memory footprint of less than 120 KB, which can easily fit into the L1/L2 cache of the chip, further accelerating memory access. Regarding real-time performance, the inference latency of 0.2039 ms is orders of magnitude lower than the typical satellite propagation delay of 20-40 ms. This implies that the security detection process

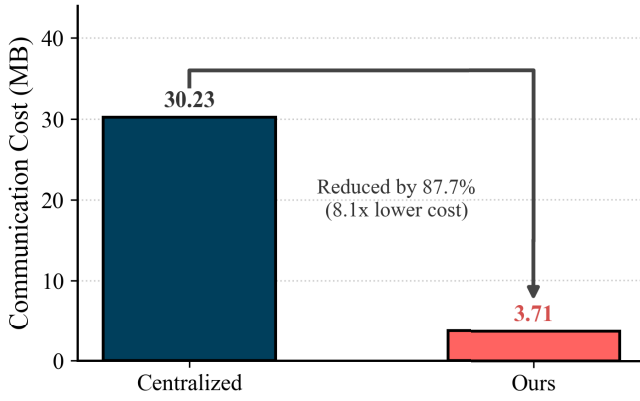


Fig. 9. Communication overhead comparison between the centralized baseline and the proposed framework.

does not introduce any perceptible jitter to the network traffic, ensuring that the intrusion detection system does not become a bottleneck for high-throughput satellite links.

2) *Communication Overhead*: As illustrated in Fig. 9, we conducted a comparative analysis of communication overhead between traditional Centralized Learning methods and the proposed SatFedGuard framework on the STI dataset. Completing the training task requires the centralized method to consume 30.23 MB of bandwidth, whereas our approach requires only 3.71 MB. This corresponds to a significant reduction in communication cost of 87.7%, translating to an $8.1\times$ improvement in communication efficiency.

The fundamental reason for this advantage lies in the inherent difference in transmission mechanisms. Centralized learning necessitates the backhaul of raw data samples from satellites to the ground station, rendering the communication overhead linearly proportional to the size of the dataset. In contrast, SatFedGuard requires only the transmission of lightweight model parameters; their size is determined by the network architecture and remains independent of the specific volume of local data. It is noteworthy that this advantage becomes increasingly pronounced as the scale of the dataset expands. In real-world LEO mega-constellation scenarios, satellites continuously generate massive volumes of traffic records. Under a centralized architecture, a surge in data volume directly results in a surge in bandwidth consumption, inevitably congesting the limited Inter-Satellite Links. However, for our Federated Learning-based approach, the communication overhead remains relatively constant, regardless of the amount of data accumulated locally. This characteristic endows our framework with exceptional scalability, making it perfectly aligned with the bandwidth-constrained environment of satellite networks.

G. Robustness and Security Analysis

1) *Resilience to Intermittent Connectivity*: To evaluate the capability of the proposed framework in handling topological instability and link failures within LEO satellite networks, we designed a simulation experiment specifically targeting intermittent connectivity. We model the availability of satellite-ground or inter-satellite links as a stochastic process.

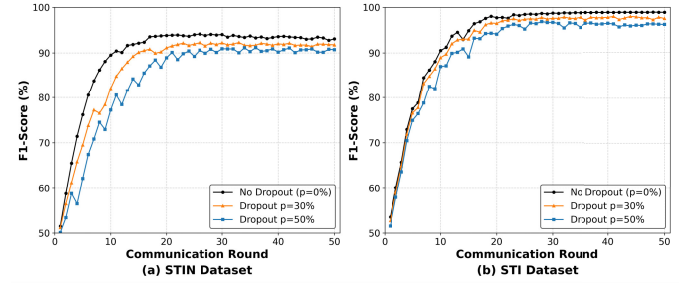


Fig. 10. Performance comparison under different intermittent connection probabilities on (a) STIN Dataset and (b) STI Dataset.

Specifically, we introduce an upload failure probability p , also referred to as the dropout rate, to represent the probability that a satellite node fails to successfully upload its local model updates during the current communication round due to factors such as physical blockage, network congestion, or link interruptions. In the experiment, we established three distinct scenarios: an ideal communication environment ($p = 0\%$), a moderate failure scenario ($p = 30\%$), and a severe failure scenario ($p = 50\%$). The experiment involved continuous observation over 50 communication rounds on both the STIN and STI datasets to record variations in the model's F1-score.

As illustrated in Fig. 10, the proposed framework exhibits exceptional robustness under intermittent connectivity. Under ideal communication conditions, the model converges rapidly. In contrast, in intermittent connectivity scenarios, the model demonstrates a noticeable lag in convergence due to the gradient sparsity inherent in single-round aggregation. Simultaneously, the stochastic nature of node participation introduces additional variance, intensifying curve fluctuations. Nevertheless, even under extremely harsh communication conditions with an upload failure probability as high as 50%—where half of the satellite nodes fail to upload updates in each round—the model achieves stable convergence without suffering from performance collapse. Specifically, on the STIN dataset, the model demonstrates superior robustness, maintaining a final F1-score of 90.7%, which represents a decline of only 2.35% compared to the ideal state. Similarly, on the STI dataset, the performance gap is merely 2.68%. These experimental results fully demonstrate the system's ability to effectively cope with the highly dynamic topological changes common in LEO mega-constellations.

This superior robustness is primarily attributed to the proposed dual-path aggregation architecture. First, in the global aggregation path, the employed FedAvg algorithm possesses inherent statistical fault tolerance; even when a subset of nodes is missing, it maintains an unbiased estimate of the global model direction by aggregating gradients from the remaining nodes, thereby ensuring fundamental convergence stability.

Second, and more critically, the architecture introduces a high-capacity Ground Station Model functioning as a teacher. Owing to its larger parameter scale and stronger feature extraction capabilities, the ground station model can extract high-level invariant features from complex raw data. Through the knowledge distillation mechanism, this essential knowledge is transferred to the lightweight satellite models. This

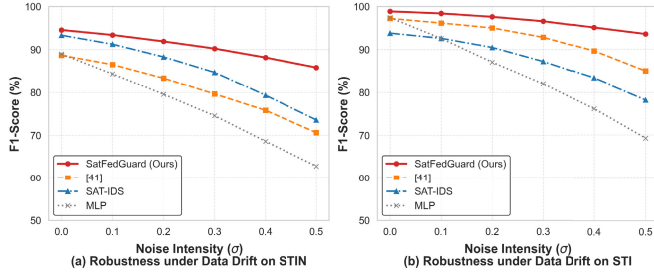


Fig. 11. Robustness evaluation under feature covariate shift on (a) STIN Dataset and (b) STI dataset.

implies that even if satellites miss portions of training data due to intermittent connectivity, the potent guidance provided by the ground station acts as a regularization constraint. This prevents the satellite models from deviating from the optimal solution, thereby significantly enhancing system robustness under unstable link conditions.

2) *Adaptability to Data Distribution Shifts*: To assess the model’s resilience against feature covariate shift—a common occurrence in dynamic satellite network environments—we simulated a data drift scenario. Specifically, Additive White Gaussian Noise (AWGN) with zero mean and standard deviation σ was injected into the feature space of the test set. The noise intensity parameter σ was varied within the range $[0.0, 0.5]$, where $\sigma = 0.0$ represents the pristine data environment. We benchmarked the proposed SatFedGuard algorithm against three baseline methods—Ref. [40], SAT-IDS, and MLP—using the F1-Score as the primary metric to observe performance trajectories under varying noise intensities.

As illustrated in Fig. 11, as the noise intensity σ increases from 0.0 to 0.5, the performance of all models exhibits a declining trend of varying degrees, as expected. This decline is attributed to the disruption of feature distributions caused by noise interference. However, significant disparities are observed in the robustness of the different models. Our method consistently maintains a leading performance position, demonstrating superior robustness. Even under severe noise conditions ($\sigma = 0.5$), it maintains F1-Scores above 85% and 93% on the STIN and STI datasets, respectively, with a degradation trajectory that is both gradual and controlled. In contrast, the baseline models exhibit heightened sensitivity to noise. Notably, MLP suffers from severe performance collapse on the STIN dataset, with its F1-Score plummeting to approximately 63%. While SAT-IDS and [40] outperform MLP, their degradation slopes are significantly steeper than that of SatFedGuard, indicating a higher susceptibility to misclassification under feature shift.

This exceptional robustness is attributed to the synergistic effect of local self-supervised learning and global knowledge guidance. On the one hand, the Barlow Twins contrastive learning algorithm deployed on satellite nodes minimizes the cross-correlation between the components of the output feature vectors. This forces the model to learn essential latent representations of the data, thereby naturally filtering out non-semantic Gaussian noise interference and rendering the model invariant to input distortions. On the other hand, this local

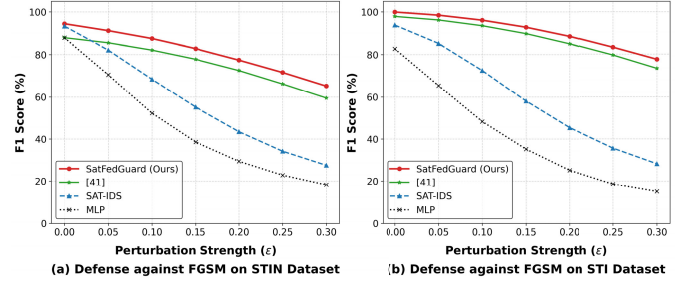


Fig. 12. Robustness evaluation against adversarial attacks on (a) STIN dataset and (b) STI dataset.

noise resilience is further reinforced by the global dual-path aggregation architecture. The soft label guidance provided by the ground station “teacher model” acts as a potent regularization term, effectively preventing satellite models from deviating from the optimal decision boundary under noise interference. Therefore, it is the combination of the feature purification capability of Barlow Twins and the deviation correction capability of knowledge distillation that establishes the leading position of SatFedGuard under feature shift conditions.

3) *Defense Against Adversarial Attack*: To evaluate the model’s security against malicious attacks, we employed the Fast Gradient Sign Method (FGSM) to simulate adversarial attack scenarios. FGSM generates adversarial examples designed to deceive the classifier by adding minute perturbations to the original input samples along the direction of the loss function’s gradient. In our experiments, we introduced a perturbation strength parameter ϵ , with values ranging from $[0.0, 0.30]$ at a step size of 0.05, where $\epsilon = 0.0$ denotes the benign environment without attacks. We benchmarked the proposed SatFedGuard framework against [40], SAT-IDS, and MLP, focusing on observing the degradation of the F1-Score for each model as the attack intensity increases, thereby quantifying the system’s defensive capabilities.

As illustrated in Fig. 12, an inevitable decline in detection performance is observed across all models as the perturbation strength ϵ increases. However, our method exhibits the most robust resilience. Experimental data indicates that on the STIN dataset, even under the most potent attack ($\epsilon = 0.30$), SatFedGuard maintains an F1-Score of 65.2%. In contrast, the traditional MLP model suffers severe performance collapse, dropping to 18.2%—a discrepancy of 47 percentage points. On the STI dataset, the advantage is equally significant: at $\epsilon = 0.30$, SatFedGuard retains a high F1-Score of 77.5%, distinctly outperforming the 73.2% of [40] and the 28.2% of SAT-IDS. Analyzing the slope of the curves, SatFedGuard displays the most gradual declining trend, indicating the lowest sensitivity to attacks.

The experimental results validate the security advantages of SatFedGuard in adversarial environments. This defensive capability is a testament to the combined effects of contrastive learning and knowledge distillation. First, the Barlow Twins contrastive learning algorithm employed by satellite nodes minimizes the cross-correlation of feature components, compelling the model to learn the intrinsic structural essence of traffic data rather than superficial features. This renders the

generated feature representations invariant to noise, making them difficult to mislead via minute gradient-based perturbations. Second, the global soft label guidance provided by the ground station teacher model acts as a potent regularization constraint. It smooths the model's decision boundaries, preventing drastic prediction deviations when the satellite model is subjected to malicious interference. Conversely, owing to the absence of such deep regularization mechanisms, MLP and SAT-IDS are highly prone to overfitting specific input patterns, rendering them extremely fragile when facing carefully crafted adversarial examples.

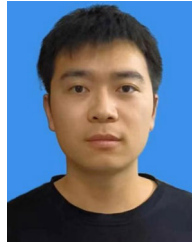
VI. CONCLUSION

This paper primarily addresses the issue of anomalous traffic detection in satellite networks when on-board labeled data is scarce. We propose SatFedGuard, a semi-supervised federated contrastive learning method for satellite network anomalous traffic detection that combines reinforcement learning-assisted personalized multi-grained bidirectional distillation. This approach enhances model performance by performing contrastive learning at each satellite node, while the ground station executes dual-path aggregation and supervised learning alongside reinforcement learning-assisted multi-grained bidirectional distillation between larger and smaller models. Experimental results demonstrate that, even with limited on-board labeled data, our method maximizes the utilization of available annotations, reducing reliance on data labels while improving on-board anomalous traffic detection performance. The F1-scores on the two datasets were boosted to 93.38% and 99.80%, respectively. In the future, we plan to focus on two aspects: first, deeply researching the dynamic topology of satellites and exploring the impact of dynamic aggregation mechanisms based on topological changes on model performance; second, we will focus on integrated space-air-ground networks and investigate collaborative learning mechanisms among cross-domain heterogeneous nodes to address the challenges posed by high heterogeneity in computational capabilities, communication conditions, and data distributions.

REFERENCES

- [1] S. Salim, N. Moustafa, and M. Reisslein, "Cybersecurity of satellite communications systems: A comprehensive survey of the space, ground, and links segments," *IEEE Commun. Surveys Tut.*, vol. 27, no. 1, pp. 372–425, Feb. 2025.
- [2] Z. Shen et al., "A survey of next-generation computing technologies in space-air-ground integrated networks," *ACM Comput. Surv.*, vol. 56, no. 1, pp. 1–40, Jan. 2024.
- [3] P. Yue et al., "Low Earth orbit satellite security and reliability: Issues, solutions, and the road ahead," *IEEE Commun. Surveys Tut.*, vol. 25, no. 3, pp. 1604–1652, 3rd Quart., 2023.
- [4] S. He, J. Ge, Y.-C. Liang, and D. Niyato, "Toward symbiotic STIN through inter-operator resource and service sharing: Joint orchestration of user association and radio resources," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 12, pp. 3674–3689, Dec. 2024.
- [5] D. Zhou, M. Sheng, Y. Wang, J. Li, and Z. Han, "Machine learning-based resource allocation in satellite networks supporting Internet of Remote Things," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6606–6621, Oct. 2021.
- [6] D. Zhou, M. Sheng, J. Li, and Z. Han, "Aerospace integrated networks innovation for empowering 6G: A survey and future challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 975–1019, 2nd Quart., 2023.
- [7] S. Mahboob and L. Liu, "Revolutionizing future connectivity: A contemporary survey on AI-empowered satellite-based non-terrestrial networks in 6G," *IEEE Commun. Surveys Tut.*, vol. 26, no. 2, pp. 1279–1321, 2nd Quart., 2024.
- [8] M. Sheng et al., "Coverage enhancement for 6G satellite-terrestrial integrated networks: Performance metrics, constellation configuration and resource allocation," *Sci. China Inf. Sci.*, vol. 66, no. 3, Mar. 2023, Art. no. 130303.
- [9] H. Al-Hraishawi, H. Chougrani, S. Kisseleff, E. Lagunas, and S. Chatzinotas, "A survey on nongeostationary satellite systems: The communication perspective," *IEEE Commun. Surveys Tut.*, vol. 25, no. 1, pp. 101–132, 1st Quart., 2023.
- [10] P. Tedeschi, S. Sciancalepore, and R. Di Pietro, "Satellite-based communications security: A survey of threats, solutions, and research challenges," *Comput. Netw.*, vol. 216, Oct. 2022, Art. no. 109246.
- [11] X. Wu et al., "Threat analysis for space information network based on network security attributes: A review," *Complex Intell. Syst.*, vol. 9, no. 3, pp. 3429–3468, Jun. 2023.
- [12] L. Fernandez, J. A. Ruiz-De-Azua, A. Calveras, and A. Camps, "On-demand satellite payload execution strategy for natural disasters monitoring using LoRa: Observation requirements and optimum medium access layer mechanisms," *Remote Sens.*, vol. 13, no. 19, p. 4014, Oct. 2021.
- [13] L. Zong, H. Wang, Y. Bai, and G. Luo, "Cross-regional transmission control for satellite network-assisted vehicular ad hoc networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 9692–9701, Jul. 2022.
- [14] S. Ma, Y. C. Chou, H. Zhao, L. Chen, X. Ma, and J. Liu, "Network characteristics of LEO satellite constellations: A starlink-based measurement from end users," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, May 2023, pp. 1–10.
- [15] L. Zhang and Y. Du, "A dynamic cascading failure model for LEO satellite networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 2, pp. 1672–1689, Apr. 2024.
- [16] Y. Liu, L. Wang, Z. Lu, and G. Shou, "A QoS guaranteed efficient integration of UPF and LEO satellite networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 4, pp. 3727–3739, Aug. 2024.
- [17] K. Li, H. Zhou, Z. Tu, W. Wang, and H. Zhang, "Distributed network intrusion detection system in satellite-terrestrial integrated networks using federated learning," *IEEE Access*, vol. 8, pp. 214852–214865, 2020.
- [18] Z. Han, C. Xu, G. Zhao, S. Wang, K. Cheng, and S. Yu, "Time-varying topology model for dynamic routing in LEO satellite constellation networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 3440–3454, Mar. 2023.
- [19] L. Zhang, Y. Du, and A. Li, "Rapid cascading risk assessment and vulnerable satellite identification schemes for LEO satellite networks," *Rel. Eng. Syst. Saf.*, vol. 256, Apr. 2025, Art. no. 110699.
- [20] S. Salim, N. Moustafa, M. Hassanian, D. Ormod, and J. Slay, "Deep-federated-learning-based threat detection model for extreme satellite communications," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 3853–3867, Feb. 2024.
- [21] Y. Zhang et al., "Security performance analysis of LEO satellite constellation networks under DDoS attack," *Sensors*, vol. 22, no. 19, p. 7286, Sep. 2022.
- [22] L. Luo, C. Zhang, H. Yu, Z. Li, G. Sun, and S. Luo, "Energy-efficient hierarchical collaborative learning over LEO satellite constellations," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 12, pp. 3366–3379, Dec. 2024.
- [23] M. Elmahallawy, T. Luo, and K. Ramadan, "Communication-efficient federated learning for LEO constellations integrated with HAPs using hybrid NOMA-OFDM," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1097–1114, May 2024.
- [24] N. Razmi, B. Matthiesen, A. Dekorsy, and P. Popovski, "On-board federated learning for satellite clusters with inter-satellite links," *IEEE Trans. Commun.*, vol. 72, no. 6, pp. 3408–3424, Jun. 2024.
- [25] Q. Wu et al., "HiFlash: Communication-efficient hierarchical federated learning with adaptive staleness control and heterogeneity-aware client-edge association," *IEEE Trans. Parallel Distrib. Syst.*, vol. 34, no. 5, pp. 1560–1579, May 2023.
- [26] M. Wan et al., "A federated learning-based intrusion detection system for satellite-terrestrial integrated networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2025, pp. 1–5.
- [27] T. Wei and K. Gan, "Towards realistic long-tailed semi-supervised learning: Consistency is all you need," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3469–3478.
- [28] Y. Fan, A. Kukleva, D. Dai, and B. Schiele, "Revisiting consistency regularization for semi-supervised learning," *Int. J. Comput. Vis.*, vol. 131, no. 3, pp. 626–643, Mar. 2023.

- [29] L.-Z. Guo, Y.-F. Li, J.-J. Shao, Z.-F. Wu, and Y.-G. Zhang, "Robust semi-supervised learning when not all classes have labels," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 3305–3317.
- [30] F. Pacheco, E. Exposito, and M. Gineste, "A framework to classify heterogeneous Internet traffic with machine learning and deep learning techniques for satellite communications," *Comput. Netw.*, vol. 173, May 2020, Art. no. 107213.
- [31] A. T. Azar, E. Shehab, A. M. Mattar, I. A. Hameed, and S. A. Elsaid, "Deep learning based hybrid intrusion detection systems to protect satellite networks," *J. Netw. Syst. Manage.*, vol. 31, no. 4, p. 82, Oct. 2023.
- [32] O. Driouch, S. Bah, and Z. Guennoun, "CANSat-IDS: An adaptive distributed intrusion detection system for satellites, based on combined classification of CAN traffic," *Comput. Secur.*, vol. 146, Nov. 2024, Art. no. 104033.
- [33] H. Xu, S. Han, X. Li, and Z. Han, "Anomaly traffic detection based on communication-efficient federated learning in space-air-ground integration network," *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 9346–9360, Dec. 2023.
- [34] J. He, X. Li, X. Zhang, W. Niu, and F. Li, "A synthetic data-assisted satellite terrestrial integrated network intrusion detection framework," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 1739–1754, 2025.
- [35] J. Jiang, B. Wang, Q. Tang, G. Zhong, X. Tang, and J. J. P. C. Rodrigues, "Incremental semi-supervised learning for data streams classification in Internet of Things," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 3, pp. 2489–2501, Jun. 2025.
- [36] Z. Dang, Y. Zheng, X. Lian, C. Peng, Q. Chen, and X. Gao, "Semi-supervised learning for anomaly traffic detection via bidirectional normalizing flows," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 5, pp. 5106–5117, Oct. 2025.
- [37] X. Zhao, M. He, and X. Wang, "Semi-supervised learning with interpolation and pseudo-labeling for few-label intrusion detection," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 5, pp. 4302–4320, Oct. 2025.
- [38] R. Zhao, Y. Wang, Z. Xue, T. Ohtsuki, B. Adebisi, and G. Gui, "Semisupervised federated-learning-based intrusion detection method for Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 8645–8657, May 2023.
- [39] Z. Wang, Z. Li, M. Fu, Y. Ye, and P. Wang, "Network traffic classification based on federated semi-supervised learning," *J. Syst. Archit.*, vol. 149, Apr. 2024, Art. no. 103091.
- [40] O. Aouedi, K. Piamrat, G. Müller, and K. Singh, "Federated semisupervised learning for attack detection in industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 19, no. 1, pp. 286–295, Jan. 2023.
- [41] Y. Li and Y. Li, "Semi-supervised federated learning for collaborative security threat detection in control system for distributed power generation," *Eng. Appl. Artif. Intell.*, vol. 148, May 2025, Art. no. 110374.
- [42] H. Hu, Y. Lyu, K. Song, R. Fan, C. Zhan, and J. Yang, "An efficient two-stage networking topology design for mega-constellation of low Earth orbit satellites," *IEEE Trans. Mobile Comput.*, vol. 24, no. 7, pp. 6333–6347, Jul. 2025.
- [43] J. Yang, B. Li, K. Fan, L. An, and Q. Zhang, "Analysis of laser inter-satellite links and topology design for mega-constellation networks," *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34554–34566, Nov. 2024.
- [44] S. Ji, D. Zhou, M. Sheng, D. Chen, L. Liu, and Z. Han, "Mega satellite constellations analysis regarding handover: Can constellation scale continue growing?," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2022, pp. 4697–4702.
- [45] G. Huang, S. Liu, L. Van der Maaten, and K. Q. Weinberger, "CondenseNet: An efficient DenseNet using learned group convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2752–2761.
- [46] Z. Cai, A. Ravichandran, S. Maji, C. Fowlkes, Z. Tu, and S. Soatto, "Exponential moving average normalization for self-supervised and semi-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 194–203.
- [47] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
- [48] V. Vapnik, S. E. Golowich, and A. Smola, "Support vector method for function approximation, regression estimation and signal processing," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 9, 1996, pp. 281–287.
- [49] T. M. Cover and P. E. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory*, vol. IT-13, no. 1, pp. 21–27, Jan. 1967.
- [50] I. Rish, "An empirical study of the naive Bayes classifier," in *Proc. IJCAI Workshop Empirical Methods Artif. Intell.*, 2001, vol. 3, no. 22, pp. 41–46.
- [51] M. Riedmiller and A. Lermen, "Multi layer perceptron," in *Machine Learning Lab Special Lecture*, vol. 24. Freiburg im Breisgau, Germany: Univ. Freiburg, 2014.



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